

# Pricing High-Dimensional American Options

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## Abstract

We discuss ...

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## 1 Introduction

We discuss ...

## 2 Information and Martingales

**Filtration:** A filtration  $\mathcal{F}_t^W$  denotes the information generated by  $W$  on the interval  $[0, t]$ . If, based on the observation of the trajectory  $\{W(s), m \geq s \geq t\}$  it is possible to decide if an event  $A \in \mathcal{F}$  has occurred or not, then  $A \in \mathcal{F}_t^W$ , likewise if the value of a random variable can be completely determined by the observations  $\{W(s), m \geq s \geq t\}$  then  $Z \in \mathcal{F}_t^W$ .

**Adaptedness:** A stochastic process  $g$  is called adapted to the filtration  $\{\mathcal{F}_t^W\}_{t \geq 0}$  if  $g(t) \in \mathcal{F}_t^W$  for all  $t \geq 0$ .

**Martingale:** A stochastic process  $\{X_t\}_{t \geq 0}$  is a Martingale w.r.t a filtration  $\{\mathcal{H}_t\}_{t \geq 0}$  if:

1.  $X_t$  is  $\mathcal{H}_t$ -measurable for all  $t$
2.  $\mathbb{E}[|X_t|] < \infty$  for all  $t$ , and
3.  $E[X_s | \mathcal{H}_t] = X_t$ , for all  $s \geq t$ .

Intuitive understanding of a Martingale is that the expectation of a future value of  $X$ , given the information available today, equals today's observed value of  $X$ . In other words, a martingale has no systematic drift.

### 3 Brownian Motion

#### 3.1 Definition of Brownian Motion:

A stochastic process  $X(t)$  is a diffusion if its local dynamics can be approximated by a stochastic differential equation (SDE) of the form:

$$X(t + \delta t) - X(t) = \mu(t, X(t))\delta t + \sigma(t, X(t))Z(t) \quad (1)$$

Where  $Z(t)$  is a normally distributed disturbance independent of everything that is happening up to time  $t$  and  $\mu(t, X(t))$  and  $\sigma(t, X(t))$  are deterministic functions.

A stochastic process is defined to be a **Wiener process**  $W_t = W(t)$  if the following conditions hold:

1. with probability 1, the mapping  $t \rightarrow W(t)$  is continuous and  $W(0) = 0$
2.  $W$  has independent increments, that is, if  $r < s \leq t < u$  then  $W(u) - W(t)$  and  $W(s) - W(r)$  are independent stochastic variables
3. for  $s < t$ :  $W(t) - W(s) \sim \mathcal{N}(0, \sqrt{t-s})$

#### 3.2 Total Variation of Brownian Motion:

The Wiener process has unbounded total variation. Consider a uniform mesh  $\Delta t = \frac{T}{N}$  then:

$$\begin{aligned} \mathbb{E} \left[ \sum_{i=0}^{N-1} |\Delta W_i| \right] &= \sum_{i=0}^{N-1} \mathbb{E} [|\Delta W_i|] = \sum_{i=0}^{N-1} \sqrt{\frac{2\Delta t_i}{\pi}} \\ &= \sqrt{\frac{2T}{\pi}} \sum_{i=0}^{N-1} \sqrt{1/N} = \sqrt{\frac{2NT}{\pi}} \rightarrow \infty, \quad \text{as } N \rightarrow \infty. \\ \mathbb{E} \left[ \int_0^T |dW(s)| \right] &= +\infty \end{aligned} \quad (2)$$

#### 3.3 Quadratic Variation of Brownian Motion:

**Partition:** A partition of a time interval  $[0, t]$  is a set of the form:  $\Pi = t_0 = 0 < t_1 < \dots < t_{n-1} = t$ .

**Size of a partition:** The size of a partition is the largest interval of the partition given by:  $||\Pi|| = \max_{0 \leq i \leq n-1} (t_{i+1} - t_i)$ .

**Quadratic Variation (QV):** The quadratic variation of a stochastic process  $X$  over a fixed time interval  $[0, t]$  is:  $[X, X]_t = \lim_{||\Pi|| \rightarrow 0} \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})^2$ , if this limit exists and does not depend on the choice of the sequence of partitions  $\Pi$ .

**Sampled QV for Brownian Motion:** Let  $W$  denote standard Brownian motion. We define  $[W, W]_t^n = \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2$  to be the sampled quadratic variation for a single partition  $\Pi$ .

**Theorem 1.** *Let  $W$  denote Brownian Motion. Then  $[W, W]_T = T$  for all  $T \geq 0$  almost surely.*

Steps for the proof:

1. Prove first that the expected value of the sampled quadratic variation  $\mathbb{E}[[W, W]_t^n] = t$ . Trivially, this implies that  $\lim_{n \rightarrow \infty} \mathbb{E}[[W, W]_t^n] = t$  as the expected value does not depend on  $n$ . Further, this result is independent of the chosen partition. Prove second that the variance of the sampled quadratic variation  $\mathbb{E}([W, W]_t^n)^2 - \mathbb{E}[[W, W]_t^n]^2 = 2\Pi t \rightarrow 0$ ,  $\lim ||\Pi|| \rightarrow 0$ . Together these results prove convergence in the mean-squared limit, in  $\mathcal{L}^2$ .
2. Now we define a standard normal random variable  $Y_{i+1} = \frac{W_{t_{i+1}} - W_{t_i}}{\sqrt{t_{i+1} - t_i}}$ , where we note that the distribution on both sides is the same regardless of the time interval. Choosing  $t_{i+1} - t_i = \frac{T}{n}$  and using the law of large numbers:

$$\frac{1}{n} \sum_{i=0}^{n-1} Y_{i+1}^2 \rightarrow \mathbb{E}[Y_{i+1}^2] = 1, \quad n \rightarrow \infty \quad (3)$$

Thus, as  $n \rightarrow \infty : \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2 = T$ .

This entails that the quadratic variation of Brownian motion behaves like a deterministic function. Almost surely means that there are some paths of Brownian motion for which the quadratic variation is not  $T$  but the probability of the set of paths for which it is not is zero; the set of paths have measure zero.

## 4 Stochastic Integrals

The aim of this section is to develop a definition of the stochastic integral  $\int_0^T f(t, W_t)$ , where  $W(t)$  is the Wiener process. We first consider some different discretisation schemes; prove the key convergence result for the Forward Euler Approximation and state the basic properties of Itô integrals.

### 4.1 Discretisation schemes for Stochastic Integrals:

The most common discretisation schemes for stochastic integrals are:

1. **Forward Euler:**  $\sum_{n=0}^{N-1} f(t_n, W(t_n))(W(t_{n+1}) - W(t_n))$
2. **Backward Euler:**  $\sum_{n=0}^{N-1} f(t_{n+1}, W(t_{n+1}))(W(t_{n+1}) - W(t_n))$
3. **Trapezoidal Method:**  $\sum_{n=0}^{N-1} \frac{(f(t_{n+1}, W(t_{n+1})) - f(t_n, W(t_n)))}{2} (W(t_{n+1}) - W(t_n))$

The limits of the forward Euler defines the Itô integral and the trapezoidal method defines the Stratonovich integral. To motivate why stochastic integrals are different from ordinary integrals we can consider the expectation values under the different discretisation schemes of the simple stochastic integral:  $\int_0^T W(t) dW(t)$ . Applying a discretisation scheme and then using independence of increments we see that each discretisation scheme yields a different result. This means we need more information to define a stochastic integral than to define a deterministic integral: we need to decide which discretisation method our solution is limit of.

### 4.2 Strong Convergence of the Forward Euler Approximation:

We will now show that if we choose the Forward Euler discretisation then discretisations will converge in  $\mathcal{L}^2$ .

#### Strong Convergence of Itô Integrals

**Theorem 2.** Suppose there exist a positive constant  $C$  such that  $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$  satisfies

$$|f(t + \Delta t, W + \Delta W) - f(t, W)| \leq C(\Delta t + |\Delta W|). \quad (2.3)$$

Consider two different partitions of the time interval  $[0, T]$

$$\{t_n\}_{n=0}^{\bar{N}}, \quad \bar{t}_0 = 0, \quad \bar{t}_{\bar{N}} = T,$$

$$\{\bar{t}_m\}_{m=0}^{\bar{N}}, \quad \bar{t}_0 = 0, \quad \bar{t}_{\bar{N}} = T,$$

with the corresponding forward Euler approximations

$$\bar{I} = \sum_{n=0}^{\bar{N}-1} f(t_n, W(t_n))(W(t_{n+1}) - W(t_n)), \quad (2.4)$$

$$\bar{\bar{I}} = \sum_{m=0}^{\bar{N}-1} f(\bar{t}_m, W(\bar{t}_m))(W(\bar{t}_{m+1}) - W(\bar{t}_m)). \quad (2.5)$$

Let the maximum time step  $\Delta t_{\max}$  be

$$\Delta t_{\max} = \max \left[ \max_{0 \leq n \leq \bar{N}-1} t_{n+1} - t_n, \quad \max_{0 \leq m \leq \bar{N}-1} \bar{t}_{m+1} - \bar{t}_m \right].$$

Then

$$\mathbb{E}[(\bar{I} - \bar{\bar{I}})^2] = \mathcal{O}(\Delta t_{\max}). \quad (4)$$

**Proof.** It is useful to introduce the finer grid made of the union of the nodes on the two grids

$$\{t_k\} \equiv \{\bar{t}_n\} \cup \{\bar{\bar{t}}_m\}.$$

Then in that grid we can write

$$\bar{I} - \bar{\bar{I}} = \sum_k \Delta f_k \Delta W_k,$$

where  $\Delta f_k = f(\bar{t}_n, W(\bar{t}_n)) - f(\bar{\bar{t}}_m, W(\bar{\bar{t}}_m))$ ,  $\Delta W_k = W(t_{k+1}) - W(t_k)$ , and the indices  $m, n$  satisfy  $t_k \in [\bar{\bar{t}}_m, \bar{\bar{t}}_{m+1})$  and  $t_k \in [\bar{t}_n, \bar{t}_{n+1})$ , as depicted in Figure 2.1.

Therefore,

$$\begin{aligned} \mathbb{E}[(\bar{I} - \bar{\bar{I}})^2] &= \mathbb{E} \left[ \sum_{k,l} \Delta f_k \Delta f_l \Delta W_l \Delta W_k \right] \\ &= 2 \sum_{k>l} \underbrace{\mathbb{E}[\Delta f_k \Delta f_l \Delta W_l \Delta W_k]}_{=\mathbb{E}[\Delta f_k \Delta f_l \Delta W_l] \mathbb{E}[\Delta W_k]=0} + \sum_k \mathbb{E}[(\Delta f_k)^2 (\Delta W_k)^2] \\ &= \sum_k \mathbb{E}[(\Delta f_k)^2] \mathbb{E}[(\Delta W_k)^2] = \sum_k \mathbb{E}[(\Delta f_k)^2] \Delta t_k. \end{aligned} \quad (2.7)$$

Using that  $|\Delta f_k|^2 \leq 2C^2(\Delta t')^2 j + (\Delta' W_k)^2$  where  $\Delta' t_k = \bar{t}_n - \bar{\bar{t}}_m \leq \Delta t_{\max}$  and  $\Delta' W_k = W(\bar{t}_n) - W(\bar{\bar{t}}_m)$  we obtain that:

$$\begin{aligned} \mathbb{E}[(\bar{I} - \bar{\bar{I}})^2] &\leq \sum_k 2C^2 \left( (\Delta' t_k)^2 + \underbrace{\mathbb{E}[(\Delta' W_k)^2]}_{=\Delta' t_k} \right) \Delta t_k \\ &\leq 2C^2 T \left( \Delta t_{\max}^2 + \Delta t_{\max} \right). \end{aligned}$$

Therefore, the sequence of approximation  $I_{\Delta T}$  is a Cauchy sequence in the Hilbert space of random variables generated by the norm  $\|I_{\Delta t}\|_{L^2} \equiv \sqrt{\mathbb{E}[I_{\Delta t}^2]}$  and the scalar product  $(X, Y) \equiv \mathbb{E}[XY]$ . We see that when  $f$  depends on the random variable  $W$  we obtain only one half order convergence. The limit of this Cauchy

sequence defines the Itô integral:

$$\sum_i f_i \Delta W_i \xrightarrow{L^2} I \equiv \int_0^T f(s, W(s)) dW(s).$$

One can prove the same for multi-dimensional Itô integrals.

### 4.3 Important properties of Itô Integrals

**Theorem 3** (Basic properties of Itô integrals). *Suppose that  $f, g: [0, T] \times \Omega \rightarrow \mathbb{R}$  are Itô-integrable (adapted and satisfying  $\sqrt{\mathbb{E}[|f(t + \Delta t, w) - f(t, w)|^2]} \leq C\sqrt{\Delta t}$  and let  $c_1, c_2 \in \mathbb{R}$  be constants. Then*

*[(i)]*

1. **Linearity**

$$\int_0^T (c_1 f(s, \cdot) + c_2 g(s, \cdot)) dW(s) = c_1 \int_0^T f(s, \cdot) dW(s) + c_2 \int_0^T g(s, \cdot) dW(s).$$

2. **Martingale property**

$$\mathbb{E} \left[ \int_0^T f(s, \cdot) dW(s) \right] = 0.$$

3. **Itô isometry**

$$\mathbb{E} \left[ \left( \int_0^T f(s, \cdot) dW(s) \right) \left( \int_0^T g(s, \cdot) dW(s) \right) \right] = \int_0^T \mathbb{E}[f(s, \cdot) g(s, \cdot)] ds.$$

## 5 Stochastic Differential Equations and Itô Calculus

In this section, we extend the convergence for Itô integrals to stochastic differential equations.

### 5.1 Strong convergence result for Itô SDEs

#### Strong Convergence of the Itô Differential equation

Let  $\bar{X}$  and  $\tilde{X}$  be forward Euler approximations of the stochastic process  $X : [0, T] \times \Omega \rightarrow \mathbb{R}$ , satisfying the stochastic differential equation

$$dX(t) = a(t, X(t))dt + b(t, X(t))dW(t), \quad 0 \leq t < T, \quad (3.4)$$

with time steps

$$\{\bar{t}_n\}_{n=0}^{\bar{N}}, \quad \bar{t}_0 = 0, \bar{t}_{\bar{N}} = T, \quad \{\tilde{t}_m\}_{m=0}^{\tilde{N}}, \quad \tilde{t}_0 = 0, \tilde{t}_{\tilde{N}} = T,$$

respectively, and

$$\Delta t_{\max} = \max \left[ \max_{0 \leq n \leq \bar{N}-1} (\bar{t}_{n+1} - \bar{t}_n), \max_{0 \leq m \leq \tilde{N}-1} (\tilde{t}_{m+1} - \tilde{t}_m) \right].$$

Suppose that there exists a positive constant  $C$  such that the initial data and the given functions  $a, b : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$  satisfy

$$E[|\bar{X}(0)|^2 + |\tilde{X}(0)|^2] \leq C, \quad (3.5)$$

$$E \left[ \left( \bar{X}(0) - \tilde{X}(0) \right)^2 \right] \leq C \Delta t_{\max}, \quad (5)$$

and

$$|a(t, x) - a(t, y)| < C|x - y|, \quad |b(t, x) - b(t, y)| < C|x - y|, \quad (3.7)$$

$$|a(t, x) - a(s, x)| + |b(t, x) - b(s, x)| \leq C(1 + |x|)\sqrt{|t - s|}. \quad (3.8)$$

Then there is a constant  $K$  such that

$$\max\{E[|\bar{X}^2(t, \cdot)|], E[|\tilde{X}^2(t, \cdot)|]\} \leq K(T + 1), \quad t < T, \quad (6)$$

and

$$E\left[\left(\bar{X}(t, \cdot) - \tilde{X}(t, \cdot)\right)^2\right] \leq K\Delta t_{\max}, \quad t < T. \quad (7)$$

An important corollary of this theorem is that this implies convergence of Itô SDE in the  $\mathbb{L}^2$  norm:  $\|\bar{X} - \tilde{X}\| = \mathcal{O}(\sqrt{\Delta t_{\max}})$ . This implies that successive refinements of the Euler approximation forms a Cauchy sequence in the Hilbert space  $V$  defined by:

$$V = \left\{ f : [0, T] \times \Omega \rightarrow \mathbb{R}^l : f \text{ is adapted and } \int_0^T \mathbb{E}[|f(t)|^2] dt < \infty \right\}$$

## 5.2 Itô formula:

**Theorem 4.** Suppose that the assumptions in Theorem 3.1 hold and that  $X$  satisfies the stochastic differential equation

$$\begin{aligned} dX(s) &= a(s, X(s))ds + b(s, X(s))dW(s), \quad s > 0 \\ X(0) &= X_0, \end{aligned}$$

and let  $g : (0, +\infty) \times \mathbb{R} \rightarrow \mathbb{R}$  be a given bounded function in  $C^2((0, \infty) \times \mathbb{R})$ . Then

$$y(t) \equiv g(t, X(t))$$

satisfies the stochastic differential equation

$$\begin{aligned} dy(t) &= \left( \partial_t g(t, X(t)) + a(t, X(t))\partial_x g(t, X(t)) + \frac{b^2(t, X(t))}{2}\partial_{xx} g(t, X(t)) \right) dt \\ &\quad + b(t, X(t))\partial_x g(t, X(t))dW(t), \end{aligned}$$

We give the steps for the proof of the Itô formula in the integral sense.

**Proof:**

1. Begin with a forward Euler approximation of  $X(t)$  the underlying process  $\bar{X}(t_n)$ .
2. Taylor expand  $g$  up to second order and then sum over time steps.
3. We then want to show that each term converges to the corresponding term in 4, using strong convergence of Itô integrals. For example:

$$Y \equiv \frac{1}{2} \sum_{n=0}^{m-1} (\bar{b})^2 \partial_{xx} g ((\Delta W_n)^2 - \Delta t_n) \rightarrow 0,$$

We can show this by show that the expectation and variance of  $Y$  go to zero.

**Quick expression for deriving Itô integrals:**

$$dg = \partial_t g dt + \partial_x g dX + \frac{1}{2} \partial_{xx} g (dX)^2$$

and the relations:  $dt dt = dt dW = dW dt = 0$  and  $dW dW = dt$ .



### 5.3 Systems of SDEs:

## 6 Black-Scholes and Feynman-Kac

### 6.1 Feynman-Kac:

We follow [3]. Consider the following Cauchy problem where we have three scalar functions  $\mu(t, x)$ ,  $\sigma(t, x)$  and  $\Phi(t, x)$  and we want find a function  $F$  which satisfies the following boundary value problem on the domain  $\mathcal{D} = [0, T] \times \mathbb{R}$

$$\partial_t F(t, x) + \mu(t, x) \partial_x F(t, x) + \frac{1}{2} \sigma(t, x) \partial_{xx}^2 F(t, x) = \partial_t F(t, x) + \hat{L}F(t, x) = 0 \quad (8)$$

$$F(t, x) = \Phi(x) \quad (9)$$

where we have defined the linear operator  $\hat{L}$ . We now aim to produce a stochastic representation formula which gives the solution to the boundary value problem in terms of the solution to an SDE. We do so by fixing a point in time and space  $(t, x)$  on the time interval  $[t, T]$  as the solution to an SDE:

$$dX_s = \mu(s, X(s))ds + \sigma(s, X(s))dW_s \quad (10)$$

$$X_t = x \quad (11)$$

Applying the Itô formula to  $F(s, X(s))$ :

$$F(T, X_T) = F(t, X_t) + \int_t^T ds \partial_s F(s, x) + \hat{L}F(s, x) + \int_t^T dW_s \sigma(s, X(s)) \partial_x F(s, X_s) \quad (12)$$

Taking the expectation value conditioning implicitly on  $X_t = x$  we obtain the Feynman-Kac representation formula:

$$\mathbb{E}[F(T, X_T)] = \mathbb{E}[F(t, X_t)] = F(t, x) \quad (13)$$

Here, we need to assume that  $\sigma(s, X_s) \partial_x F(s, X_s)$  is in  $\mathcal{L}^2$  to ensure that the expected value of the stochastic integral equals to zero.

### 6.2 Kolmogorov Equations:

Suppose  $X$  is a solution to 10 and 11 with our generator given by the operator  $\hat{L}$ . Consider the boundary value problem:

$$(\partial_s u + \hat{L}u)(s, y) = 0, \quad (s, y) \in (0, T] \times \mathbb{R} \quad (14)$$

$$u(T, y) = \mathbb{1}_B(y), \quad y \in \mathbb{R} \quad (15)$$

where  $\mathbb{1}_B(y)$  is an indicator function of the set  $B$ . Then by Feynman-Kac we have:

$$u(s, y) = \mathbb{E}_{s,y}[\mathbb{1}_B(X_T) | X_s = y] = P(X_T \in B | X_s = y) \quad (16)$$

Hence, we obtain the **Kolmogorov Backward Equation** (KBE) for the transition probabilities which satisfy:

$$(\partial_s P + \hat{L}P)(s, y; t, B) = 0, \quad (s, y) \in (0, T] \times \mathbb{R} \quad (17)$$

$$P(t, y; t, B) = \mathbb{1}_B(y) \quad (18)$$

An analogous equation holds for the transition probability densities given by:

$$(\partial_s p + \hat{L}p)(s, y; t, B) = 0, \quad (s, y) \in (0, T] \times \mathbb{R} \quad (19)$$

$$p(t, y; t, x) \rightarrow \delta(x - y) \quad \text{as } s \rightarrow t \quad (20)$$

We may derive the **Kolmogorov Forward Equation** also known as the **Fokker-Planck Equation** by fixing two points in time  $(s, T)$  with  $s < T$  considering an arbitrary test function i.e. an infinite differentiable function  $h(t, x)$  with compact support in the set  $(s, T) \times \mathbb{R}$ . By the Itô formula and assuming that  $h(t, x)$  has compact support we obtain that:

$$\int_s^T (\partial_t h + \hat{L}h)(t, X_t) dt + \int_s^T \partial_x h(t, X_t) dW_t \quad (21)$$

Taking expectation values, integrating over the probability density and integrating by parts we obtain:

$$\int_{-\infty}^{\infty} dx \int_t^T dt h(-\partial_t p + \hat{L}^* p)(s, y; x, t) \quad (22)$$

with operator  $\hat{L}^*$  given by:

$$(\hat{L}^* f)(t, x) = -\partial_x(\mu(t, x)f(t, x)) + \frac{1}{2}\partial_{xx}^2(\sigma(t, x)f(t, x)) \quad (23)$$

Since this must hold for an arbitrary test function we obtain the Fokker-Planck equation:

$$(-\partial_t p + \hat{L}^* p)(s, y; x, t) = 0 \quad (24)$$

### 6.3 Black-Scholes Equation:

Let  $f(t, S_t)$  be the price of a European option written on a stock  $S(t)$ , which we assume evolves according to Geometric Brownian Motion (GBM):  $dS_t = \mu S_t dt + \sigma S_t dW_t$  where the drift  $\mu$  and volatility  $\sigma$  are constants. We also assume the existence of a riskless bond which follows  $dB = rBdt$ , where  $r$  is the risk-free rate, also assumed constant. We aim to derive the PDE satisfied by  $f(t, S_t)$ .

We begin by considering the portfolio  $I = -f + \alpha S + \beta B$ , for  $\alpha(t), \beta(t) \in \mathbb{R}$ . Assuming that this portfolio is self-financing then  $dI = -df + \alpha dS + \beta dB$  and Itô's formula:

$$\begin{aligned} dI &= -df + \alpha dS + \beta dB \\ &= -\left(f_t + \mu S f_s + \frac{1}{2}\sigma^2 S^2 f_{ss}\right) dt - f_s \sigma S dW + \alpha(\mu S dt + \sigma S dW) + \beta r B dt \\ &= \left(-\left(f_t + \mu S f_s + \frac{1}{2}\sigma^2 S^2 f_{ss}\right) + (\alpha\mu S + \beta r B)\right) dt + (-f_s + \alpha)\sigma S dW. \end{aligned}$$

Now choose  $\alpha$  such that the portfolio  $I$  becomes riskless, i.e.  $\alpha = f_s$ , so that

$$\begin{aligned} dI &= \left(-\left(f_t + \mu S f_s + \frac{1}{2}\sigma^2 S^2 f_{ss}\right) + (f_s \mu S + \beta r B)\right) dt \\ &= \left(-\left(f_t + \frac{1}{2}\sigma^2 S^2 f_{ss}\right) + \beta r B\right) dt. \end{aligned}$$

Assume also that the existence of an arbitrage opportunity is precluded, i.e.  $dI = rIdt$ , where  $r$  is the interest rate for riskless investments, to obtain

$$\begin{aligned} dI &= r(-f + \alpha S + \beta B)dt \\ &= r(-f + f_s S + \beta B)dt. \end{aligned}$$

Equation (above) shows that

$$f_t + rSf_s + \frac{1}{2}\sigma^2 S^2 f_{ss} = rf, \quad t < T,$$

In this derivation, we have assumed that  $r, \mu, \sigma$  are all constant, however, the same equation holds if they depend on time and stock price. Further, we note that the Black-Scholes equation does not depend on the local mean rate of return of the underlying asset  $\mu$  so Black-Scholes is blind to the rate of return of the stock. The only aspect of the underlying process which has any significance is the volatility  $\sigma$ .

## 6.4 Risk-free Measure:

The Black-Scholes PDE can be solved using the Feynman-Kac representation formula  $F(t, s) = e^{r(T-t)} \mathbb{E}_{t,s}[\Phi(X(T))]$  with underlying process given by:

$$dX_t = rX_t dt + \sigma X_t dW_t \quad (25)$$

which is the same as the SDE of the stock price with  $\mu = r$ . Hence, we can define a new probability measure  $\mathbb{Q}$  under which the  $S$ -process has a different probability distribution. This measure is known as the risk-free measure and is also a martingale measure. From now on, in financial contexts, we take expectation values with respect to  $\mathbb{E}^{\mathbb{Q}}$ .

## 7 Stochastic Optimal Control:

### 7.1 Optimal control, Inverse problems and well-posedness:

Optimal control aims to control the behaviour of dynamical systems by introducing further parameters that system variables depend on. Inverse problems determine input data in an equation from its solution values. These kinds of problems are typically harder to solve numerically than forward problems because they are ill-posed. We define a problem to ill-posed if it is not well-posed, where a problem is well-posed iff:

1. there is a solution
2. the solution is unique
3. the solution depends continuously on the data

### 7.2 Definition of Stochastic Optimal Control:

We study optimal control of the solution  $X(t)$  to the stochastic differential equation

$$\begin{cases} dX_i(s) = a_i(X(s), \alpha(s, X(s)))ds \\ \quad + b_{ij}(X(s), \alpha(s, X(s)))dW_j(s), & t < s < T \\ X(t) = x \end{cases}$$

where  $T$  is a fixed terminal time and  $x \in \mathbb{R}^n$  is a given initial point. Assume that  $a_i, b_{ij} : \mathbb{R}^n \times A \rightarrow \mathbb{R}$  are smooth bounded functions, where  $A$  is a given compact subset of  $\mathbb{R}^m$ .

The function  $\alpha : [0, T] \times \mathbb{R}^n \rightarrow A$  is a *control* and let  $\mathcal{A}$  be the set of admissible Markov control functions  $t \mapsto \alpha(t, X(t))$ . These use the current value  $X(s)$  to affect the dynamics of  $X$  by adjusting the drift and the diffusion coefficients. For  $\alpha \in \mathcal{A}$  define the *cost* functional

$$C_{t,x}(\alpha) = E \left[ \int_t^T h(X(s), \alpha(s)) ds + g(X(T)) \right]$$

where  $X$  solves the stochastic differential equation above (with control  $\alpha$ ) and

$$h : \mathbb{R}^n \times A \rightarrow \mathbb{R}, \quad g : \mathbb{R}^n \rightarrow \mathbb{R}$$

are given smooth bounded functions. We call  $h$  the *running cost* and  $g$  the *terminal cost* and we aim to find the optimal control  $\alpha^*$  which minimises the expected cost  $C_{t,x}(\alpha)$ . Define the value function:

$$u(t, x) \equiv \inf_{\alpha \in \mathcal{A}} C_{t,x}(\alpha) \quad (26)$$

### 7.3 Dynamic-Programming Relation:

**Lemma 1.** *Assume that sufficient regularity assumptions hold. Then, the function  $u$  satisfies, for all  $\delta > 0$ , the dynamic programming relation*

$$u(t, x) = \min_{\alpha: [t, t+\delta] \rightarrow A} E \left[ \int_t^{t+\delta} h(X(s), \alpha(s, X(s))) ds + u(t + \delta, X(t + \delta)) \right].$$

The importance of Dynamic-Programming relation is that we no longer have to solve for  $\alpha$  over the whole interval  $[t, T]$  but only over the smaller interval  $[t, t + \delta]$  as long as the function at  $u(t + \delta, x + \delta)$  is known, decomposing a given optimal control problem into solutions with simpler problems. We can then iterate backwards to find the initial data. The proof has two steps: to use the optimal control to verify that

$$u(t, x) \geq \min_{\alpha \in A} E \left[ \int_t^{t+\delta} h(X(s), \alpha(s)) ds + u(t + \delta, X(t + \delta)) \right],$$

and then to show that an arbitrary control yields

$$u(t, x) \leq \min_{\alpha \in A} E \left[ \int_t^{t+\delta} h(X(s), \alpha(s)) ds + u(t + \delta, X(t + \delta)) \right],$$

which together imply Lemma 1.

#### Step 1:

Choose the optimal control  $\alpha^*$ , from  $t$  to  $T$ , to obtain

$$\begin{aligned} u(t, x) &= \min_{\alpha \in A} E \left[ \int_t^T h(X(s), \alpha(s, X(s))) ds + g(X(T)) \right] \\ &= E \left[ \int_t^{t+\delta} h(X(s), \alpha^*(s)) ds \right] \\ &\quad + E \left[ \int_{t+\delta}^T h(X(s), \alpha^*(s)) ds + g(X(T)) \right] \\ &= E \left[ \int_t^{t+\delta} h(X(s), \alpha^*(s)) ds \right] \\ &\quad + E \left[ E \left[ \int_{t+\delta}^T h(X(s), \alpha^*(s)) ds + g(X(T)) \mid X(t + \delta) \right] \right] \\ &\geq E \left[ \int_t^{t+\delta} h(X(s), \alpha^*(s)) ds \right] + E [u(t + \delta, X(t + \delta))] \\ &\geq \min_{\alpha \in A} E \left[ \int_t^{t+\delta} h(X(s), \alpha(s, X(s))) ds + u(t + \delta, X(t + \delta)) \right]. \end{aligned}$$

#### Step 2:

Choose the control  $\alpha^+$  to be arbitrary from  $t$  to  $t + \delta$  and then, given the value  $X(t + \delta)$ , choose the optimal  $\alpha^*$  from  $t + \delta$  to  $T$ . Denote this control by  $\alpha' = (\alpha^+, \alpha^*)$ . Definition (229) shows

$$\begin{aligned}
u(t, x) &\leq C_{t,x}(\alpha') = \mathbb{E} \left[ \int_t^T h(X(s), \alpha'(s)) ds + g(X(T)) \right] \\
&= \mathbb{E} \left[ \int_t^{t+\delta} h(X(s), \alpha^+(s)) ds \right] \\
&\quad + \mathbb{E} \left[ \int_{t+\delta}^T h(X(s), \alpha^*(s)) ds + g(X(T)) \right] \\
&= \mathbb{E} \left[ \int_t^{t+\delta} h(X(s), \alpha^+(s)) ds \right] \\
&\quad + \mathbb{E} \left[ \mathbb{E} \left[ \int_{t+\delta}^T h(X(s), \alpha^*(s)) ds + g(X(T)) \mid X(t+\delta) \right] \right] \\
&= \mathbb{E} \left[ \int_t^{t+\delta} h(X(s), \alpha^+(s)) ds \right] + \mathbb{E} [u(t+\delta, X(t+\delta))].
\end{aligned}$$

Taking the minimum over all controls  $\alpha^+$  yields

$$u(t, x) \leq \min_{\alpha^+ \in \mathcal{A}} \mathbb{E} \left[ \int_t^{t+\delta} h(X(s), \alpha^+(s)) ds + u(t+\delta, X(t+\delta)) \right].$$

We can then use the Dynamic-Programming relation to obtain the optimal control  $\alpha$  given a value function  $u$ .

Let us take the limit when  $\delta \rightarrow 0$  in 1

$$0 = \min_{\alpha: [t, t+\delta] \rightarrow A} \frac{1}{\delta} \mathbb{E} \left[ \int_t^{t+\delta} h(X(s), \alpha(s, X(s))) ds + (u(t+\delta, X(t+\delta)) - u(t, x)) \right].$$

Then, by continuity, Itô's formula and conditioning on  $X(t) = x$ , we just need to find  $\alpha = \alpha(t, x) \in A$ , a point value defined by

$$0 = \partial_t u(t, x) + \min_{\alpha \in A} \left\{ h(x, \alpha) + a_i(x, \alpha) \partial_{x_i} u(t, x) + \frac{b_{ik}(x, \alpha) b_{jk}(x, \alpha)}{2} \partial_{x_i x_j} u(t, x) \right\}.$$

so indeed we have

$$\alpha(t, x) \in \arg \min_{\alpha \in A} \left\{ h(x, \alpha) + a_i(x, \alpha) \partial_{x_i} u(t, x) + \frac{b_{ik}(x, \alpha) b_{jk}(x, \alpha)}{2} \partial_{x_i x_j} u(t, x) \right\}$$

## 7.4 Hamilton-Jacobi-Bellman equation:

**Theorem 5.** Assume that  $X$  solves (228) with a Markov control function  $\alpha$  and that the function  $u$  defined by (229) is bounded and smooth. Then  $u$  satisfies the Hamilton-Jacobi-Bellman (nonlinear PDE) equation

$$\partial_t u(t, x) + H(t, x, Du(t, x), D^2 u(t, x)) = 0, \quad t < T$$

$$u(T, x) = g(x),$$

with the Hamiltonian function (depending on the vector  $p$  and the symmetric matrix  $V$ )

$$H(t, x, p, V) \equiv \min_{\alpha \in A} \left[ a_i(x, \alpha) p_i + \frac{b_{ik}(x, \alpha) b_{jk}(x, \alpha)}{2} V_{ij} + h(x, \alpha) \right]$$

that is

$$H(t, x, Du(t, x), D^2u(t, x)) \equiv \min_{\alpha \in A} \left[ a_i(x, \alpha) \partial_{x_i} u(t, x) + \frac{b_{ik}(x, \alpha) b_{jk}(x, \alpha)}{2} \partial_{x_i x_j} u(t, x) + h(x, \alpha) \right]$$

This means that the value function is uniquely defined as the solution of the PDE is unique, although the control  $\alpha$  is not necessarily unique because of the minimisation (maximisation).

## 7.5 A simple analytically-solvable example:

## 7.6 Maximum Norm Stability of HJB:

# 8 American Options

## 8.1 European Options

## 8.2 Introduction to pricing American Options

The price of an American option  $u_A$  may be expressed as:

$$u_A(t, x) = \sup_{\tau \in \mathcal{T}_t} \mathbb{E} \left[ e^{-r(\tau-t)} g(\vec{P}_1 \cdot \vec{X}(\tau)) \mid \vec{X}(t) = \vec{x} \right], \quad (27)$$

where for any  $q \in [0, T]$  the admissible stopping times are given by

$$\mathcal{T}_q = \left\{ \tau : \Omega \rightarrow [q, T] \mid \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in [q, T] \right\}. \quad (28)$$

with the filtration  $\mathcal{F}_t = \sigma\{\vec{X}(q) : 0 \leq q \leq t\}$ . **strategy - key difference between American and European options is that it is a decision problem - you first have to decide a policy - find th optimal tau first.**

## 8.3 American Options as a Stochastic Optimal Control Problem

The American option price may be formulated as a stochastic optimal control (SOC) problem:

$$\begin{cases} dX_i(s) = \alpha(s, X(s)) a_i(X(s)) ds \\ \quad + \alpha(s, X(s)) b_{ij}(X(s)) dW_j(s), & t < s < T \\ X(t) = x \end{cases}$$

where  $T$  is a fixed terminal time and  $x \in \mathbb{R}^n$  is a given initial point. Assume that  $a_i, b_{ij} : \mathbb{R}^n \rightarrow \mathbb{R}$  are smooth bounded functions, where  $A = \{0, 1\}$ .

The corresponding Hamilton-Jacobi-Bellman (HJB) equation given by:

$$-\partial u_A(t, x) = (\mathcal{H}u_A)(t, x), \quad (t, x) \in [0, T] \times D, \quad (29)$$

$$u_A(t, \cdot) = g(\vec{P}_1 \cdot \cdot) \quad (30)$$

The Hamiltonian is written compactly as:

$$(\mathcal{H}u_A)(t, x) = \max_{\alpha \in \{0, 1\}} (\alpha \mathcal{L}u_A)(t, x) = (\mathcal{L}u_A)^+(t, x), \quad (31)$$

with the second order linear differential operator  $\mathcal{L}$  given by:

$$(\mathcal{L}v)(t, \vec{x}) = \left( -r + \sum_i a_i \partial_{x_i} + \frac{1}{2} \sum_{i,j} (\mathbf{b}\mathbf{b}^\top)_{ij} \partial_{x_i x_j}^2 \right) v(t, \vec{x}). \quad (32)$$

## 9 Markovian Projection Method

### 9.1 Markovian Projection Lemma:

Strong Convergence of the Itô Differential equation

**Lemma 2.** The time evolution of the vector of asset prices  $\vec{X} \in \mathbb{R}^d$  is given by:

$$dX_i(t) = a_i(t, X(t))dt + b_{ij}(t, X(t))dW_j(t), \quad 0 \leq t \leq T \quad (33)$$

$$\vec{X}(0) = \vec{x}_0 \quad (34)$$

with  $\vec{W}$  a k-dimensional Wiener process with independent components.

Consider the (generally) non-Markovian process  $S_1 = \vec{P}_1 \cdot \vec{X}$ ,  $\vec{P}_1 \in \mathbb{R}^{d \times 1}$  and the surrogate process  $\bar{S}^{x_0} \in \mathbb{R}$ , whose dynamics are given by:

$$d\bar{S}^{x_0}(t) = \bar{a}^{x_0}(t, \bar{S}^{x_0}(t))dt + \bar{b}^{x_0}(t, \bar{S}^{x_0}(t))dW(t), \quad t \in [0, T] \quad (35)$$

$$\bar{S}^{x_0}(0) = \vec{P}_1 \cdot \vec{x}_0 \quad (36)$$

and the diffusion coefficients  $\bar{a}^{x_0}$  and  $\bar{b}^{x_0}$  are determined as:

$$\bar{a}^{x_0}(t, s) = \mathbb{E}[\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (37)$$

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\vec{P}_1^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (38)$$

Then,  $S_1(t) |_{\vec{X}(0)=\vec{x}_0} = \vec{P}_1 \cdot \vec{X}(t) |_{\vec{X}(0)=\vec{x}_0}$  and  $\bar{S}^{x_0}(t)$  have the same conditional distribution for all  $t \in [0, T]$ .

We note that in general  $S_1(t) |_{\vec{X}(0)=\vec{x}_0} = \vec{P}_1 \cdot \vec{X}(t) |_{\vec{X}(0)=\vec{x}_0}$  and  $\bar{S}^{x_1}(t)$ , where  $\vec{x}_1 \neq \vec{x}_0$ , do not have the same conditional distribution, although we expect that for  $\vec{x}_1$  close to  $\vec{x}_0$ , the distribution conditional on  $\vec{x}_1$  to be well-approximated by the  $\vec{x}_0$  coefficients  $\bar{a}^{x_0}$  and  $\bar{b}^{x_0}$ .

**Proof:**

Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  and consider the weak approximation error

$$\epsilon_T = \mathbb{E}[g(\bar{S}^{(x_0)}(T)) | \bar{S}^{(x_0)}(0) = P_1 x_0] - \mathbb{E}[g(P_1 X(T)) | X(0) = x_0]. \quad (39)$$

Let us also define

$$\bar{u}(t, s) = \mathbb{E}[g(\bar{S}^{(x_0)}(T)) | \bar{S}^{(x_0)}(t) = s]. \quad (40)$$

Then, we represent the projection weak error as

$$-\epsilon_T = \mathbb{E}[\bar{u}(T, S_1(T)) | X(0) = x_0] - \bar{u}(0, S_1(0)) \quad (41)$$

$$= \mathbb{E} \left[ \int_0^T (\partial_t \bar{u}(t, S_1(t)) + [\mathcal{G}\bar{u}](t, X(t))) dt \right] \quad (42)$$

where the non-Markovian but Itô-integrable process  $S_1$  solves

$$dS_1(t) = P_1 a(t, X(t))dt + P_1 b(t, X(t))dW_t, \quad (43)$$

and we have the generator

$$[\mathcal{G}\bar{u}](t, x) = P_1 a(t, x) \partial_s \bar{u}(t, P_1 x) + \frac{1}{2} (P_1 b b^T P_1^T)(t, x) \partial_{ss} \bar{u}(t, P_1 x). \quad (44)$$

Using the fact that  $\bar{u}$  solves the projected KBE equation:

$$0 = \partial_t \bar{u} + \bar{\mathcal{G}}\bar{u}, \quad t < T, \quad (45)$$

$$\bar{u}(T, \cdot) = g(\cdot), \quad (46)$$

with coefficients  $\bar{a}^{(x_0)}$ ,  $\bar{b}^{(x_0)}$  and

$$\bar{\mathcal{G}}v = \bar{a}^{(x_0)} \partial_s v + \frac{1}{2} (\bar{b}^{(x_0)})^2 \partial_{ss} v,$$

we obtain

$$-\epsilon_T = \int_0^T \mathbb{E}_{0,x_0} \left[ \left( P_1 a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1 X(t)) \right) \partial_s \bar{u}(t, P_1 X(t)) \right] dt \quad (47)$$

$$+ \frac{1}{2} \int_0^T \mathbb{E}_{0,x_0} \left[ \left( (P_1 b b^\top P_1^\top)(t, X(t)) - (\bar{b}^{(x_0)})^2(t, P_1 X(t)) \right) \partial_{ss} \bar{u}(t, P_1 X(t)) \right] dt. \quad (48)$$

Let us now choose  $\bar{a}^{(x_0)}$  and  $\bar{b}^{(x_0)}$  such that  $\epsilon_T = 0$ . Using the tower property and that by conditioning on  $P_1 X(t)$  then  $\bar{u}(t, P_1 X(t))$  and  $\bar{a}^{(x_0)}(t, P_1 X(t))$  become deterministic functions so can be taken out of the inner expectation we obtain:

$$\mathbb{E}_{0,x_0} \left[ \left( P_1 a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1 X(t)) \right) \partial_s \bar{u}(t, P_1 X(t)) \right] \quad (49)$$

$$= \mathbb{E}_{0,x_0} \left[ \mathbb{E} \left[ \left( P_1 a(t, X(t)) - \bar{a}^{(x_0)}(t, P_1 X(t)) \right) \partial_s \bar{u}(t, P_1 X(t)) \mid P_1 X(t) \right] \right] \quad (50)$$

$$= \mathbb{E}_{0,x_0} \left[ \left( \mathbb{E}[P_1 a(t, X(t)) \mid P_1 X(t)] - \bar{a}^{(x_0)}(t, P_1 X(t)) \right) \partial_s \bar{u}(t, P_1 X(t)) \right]. \quad (51)$$

We note that it is precisely because the generators  $\mathcal{G}$  and  $\bar{\mathcal{G}}$  both end up depending on  $\partial_s \bar{u}_A(t, P_1 X)$  which allows the factorisation and ensures that this term can be made to vanish for all bounded and continuous functions  $g$  (as well as for all  $T$  since  $\bar{a}^{(x_0)}$  and  $\bar{b}^{(x_0)}$  does not depend on  $T$ ). A similar derivation applies to the second term. Thus, we conclude that

$$0 = \mathbb{E} \left[ g(S_1(T)) - g(\bar{S}^{(x_0)}(T)) \mid X(0) = x_0 \right] \quad (52)$$

for all bounded continuous functions  $g$ . Therefore, for all fixed  $T$ , the random variable  $S_1(T) = P_1 X(T)$  has the same conditional distribution as  $\bar{S}^{(x_0)}(T)$ .

### 9.1.1 Markovian Projection for Pricing European Options under Black-Scholes Model

### 9.1.2 Estimating the projected volatility under Black-Scholes and verifying Markovian Projection Lemma

Black-Scholes model for  $d$  uncorrelated Wiener processes is given by:

$$\begin{aligned} dX_i(t) &= rX_i dt + \sigma X_i dW_i(t) \\ \vec{X}(0) &= \vec{x}_0 \end{aligned} \quad (53)$$

We consider a basket stocks to be equal-weighted such that  $\vec{P} \cdot \vec{X} = \frac{1}{d} \sum_i X_i$ . Our non-Markovian process  $S = \vec{P} \cdot \vec{X}$  is the solution to the SDE:

$$\begin{aligned} dS &= \frac{1}{d} \sum_i rX_i dt + \sigma X_i dW_i(t) \\ S(0) &= \vec{P} \cdot \vec{x}_0 \end{aligned} \quad (54)$$

We now consider the Markovian projection given by:

$$d\bar{S}^{x_0}(t) = \bar{a}^{x_0}(t, \bar{S}^{x_0}(t))dt + \bar{b}^{x_0}(t, \bar{S}^{x_0}(t))dW(t), \quad t \in [0, T] \quad (55)$$



$$\bar{S}^{x_0}(0) = \bar{P}_1 \cdot \vec{x}_0 \quad (56)$$

Our coefficients  $\bar{a}^{x_0}$  and  $\bar{b}^{x_0}$  are then given by:

$$\begin{aligned} \bar{a}^{x_0}(t, s) &= \mathbb{E}[\vec{P} \cdot \vec{a}(t, \vec{X}(t)) | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[r \vec{P} \cdot \vec{X}(t) | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= rs \end{aligned} \quad (57)$$

$$\begin{aligned} (\bar{b}^{x_0}(t, s))^2 &= \mathbb{E}[\vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}\left[\frac{1}{d^2} \sum_i \sigma^2 X_i^2 | \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0\right] \end{aligned} \quad (58)$$

We will need to approximate  $\bar{b}^{x_0}$  and to do so we recall that for two random variable  $X$  and  $Y$ , the conditional expected value  $\mathbb{E}[X|Y] = g(Y)$  satisfies the *best mean square approximation*:

$$\mathbb{E}[(X - g(Y))^2] \leq \mathbb{E}[(X - h(Y))^2] \quad (59)$$

for all function  $h$ ,  $\mathbb{E}[h(Y)^2] < \infty$  and so given  $V = \{h : \mathbb{E}[h(Y)^2] < \infty\}$ :

$$g(Y) = \arg \min_{h \in V} \mathbb{E}[(X - h(Y))^2] \quad (60)$$

This means we can related finding an expected value to an  $\mathcal{L}^2$  regression/minimisation problem.

We model the coefficients in the time and space domain such that:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \int_0^T \mathbb{E}\left[\left(\vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} - h(t, \vec{P} \cdot \vec{X})\right)^2 | \vec{X}(0) = \vec{x}_0\right] dt \quad (61)$$

Where  $V$  will be the set of ansatz functions to model space and time. We can approximate this integral by discretising the time and applying a Monte Carlo estimate for the expected value. Assuming a grid given by  $0 = t_0 < t_1 \dots < t_N = T$  we obtain that:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M} \sum_m \frac{1}{N} \sum_n \left( \frac{1}{d^2} \sum_i \sigma^2 X_i^m(t_n)^2 - h(t, \vec{P} \cdot \vec{X}^m(t_n)) \right)^2 \quad (62)$$

where  $\vec{X}^m(t)$  are independent realisations of  $\vec{X}$ .

### Regression problem:

Assume we have the monomial basis  $(\phi_0, \phi_1, \dots)$  with  $\phi_i(x) = x^i$ . To build a basis of polynomials on  $\mathbb{R}^2$  we use:

$$\phi_{(i_1, i_2)}(t, x) = t^{i_1} x^{i_2}, \quad (i_1, i_2) \in \Lambda \quad (63)$$

The ansatz function is based on this basis:

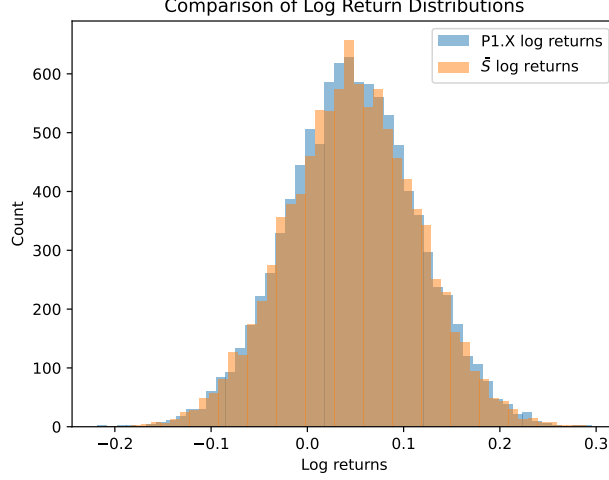
$$h(t, s) = \sum_{(i_1, i_2) \in \Lambda} c_{(i_1, i_2)} \phi_{(i_1, i_2)}(t, s) \quad (64)$$

We now want to derive the normal equation to this end we introduce notation:

$$D_{(m, n), p} = \phi_p(t_n, P X^m(t_n)), \quad m = 1, \dots, M, \quad n = 0, \dots, N-1, \quad p \in \Lambda,$$

$$\{D_{k, p}\}_{k=1, \dots, MN}^{p \in \Lambda} \in \mathbb{R}^{MN \times |\Lambda|},$$

$$\psi_k = \frac{1}{d^2} \sum_{i=1}^d \sigma^2 (X_i^m(t_n))^2, \quad \psi \in \mathbb{R}^{MN}.$$



**Figure 1:** Histogram of the log returns of a basket of 5 stocks as computed by Markov projection  $\bar{S}$ , where projected volatility was estimated using  $\mathcal{L}^2$  regression and from the true process  $\bar{P} \cdot \bar{X}$ . We observe good agreement between the distribution of the log-returns for both projected and true process.

The optimal coefficients are given by

$$\begin{aligned} \vec{c} &= \arg \min_{\{c_{i_1, i_2}\}_{(i_1, i_2) \in \Lambda}} \frac{1}{MN} \sum_{m=1}^M \sum_{n=0}^{N-1} \left( \frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^m(t_n)^2 - \sum_{(i_1, i_2) \in \Lambda} c_{i_1, i_2} \phi_{(i_1, i_2)}(t_n, PX^m(t_n)) \right)^2 \\ &= \arg \min_{\vec{c}} (\psi - D\vec{c})^\top (\psi - D\vec{c}) \end{aligned} \quad (65)$$

Equivalently, defining

$$J(\vec{c}) = \psi^\top \psi - 2\vec{c}^\top D^\top \psi + \vec{c}^\top D^\top D \vec{c},$$

we set

$$\frac{\partial J(\vec{c})}{\partial \vec{c}} = -2D^\top \psi + 2D^\top D \vec{c} = 0.$$

We implement this in Python, where we first generate  $M_t = 100$  paths of  $\bar{X}$  using the forward Euler approximation. After deriving the design matrix  $D$  and  $\psi$ , we solve the normal equation to obtain the coefficients  $\vec{c}$ . We then test the Markov projection against the true d-dimensional distribution applying  $\bar{P}$  at the final time. We note that the regression becomes more stable with an orthogonal polynomial basis with respect to the density:  $\bar{P} \cdot \bar{X}$ . Although the exact density is unknown, we can use the empirical density to derive an inner product and an orthogonal basis with respect to this inner product. Our results are shown in Fig.1 for  $10^4$  Monte Carlo paths.

## 9.2 Markovian Projection for Pricing American Options

### 9.2.1 Error bounds

We want to show that:

$$\begin{aligned} u_A(0, \vec{x}) - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}) &\leq \int_0^T \mathbb{E} \left[ \mathcal{H} \left( t, \bar{X}(t), \bar{P}_1 D \bar{u}_A(t, \bar{P}_1 \cdot \bar{X}(t)), \bar{P}_1 D^2 \bar{u}_A(t, \bar{P}_1 \cdot \bar{X}(t)) \bar{P}_1^\top \right) \right. \\ &\quad \left. - \bar{\mathcal{H}}(t, \bar{P}_1 \cdot \bar{X}(t), D \bar{u}_A(t, \bar{P}_1 \cdot \bar{X}(t)), D^2 \bar{u}_A(t, \bar{P}_1 \cdot \bar{X}(t))) \right] dt \end{aligned} \quad (66)$$

$$\begin{aligned} \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}) - u_A(0, \vec{x}) &\leq \int_0^T \mathbb{E} \left[ -H(t, \bar{S}_1(t) \vec{P}_1, Du_A(t, \bar{S}_1(t) \vec{P}_1), D^2 u_A(t, \bar{S}_1(t) \vec{P}_1)) \right. \\ &\quad \left. + \bar{H}(t, \bar{S}_1(t), \vec{P}_1 Du_A(t, \bar{S}_1(t) \vec{P}_1), \vec{P}_1^T D^2 u_A(t, \bar{S}_1(t) \vec{P}_1) \vec{P}_1) \right] dt \end{aligned} \quad (67)$$

This gives us the bounds on the error of using Markovian projection for American options. We explain below why this error arises in the first place.

Define

$$\bar{u}_A(t, s) = \mathbb{E}[g(\bar{S}(T)) | \bar{S}(t) = s] \quad (68)$$

with a control given by:  $\bar{\alpha}(t, x) = 1_{\bar{u}_A(t, P_1 x) > g(P_1 x)}$ .

$$u_A(t, \vec{x}) = \mathbb{E}[g(\vec{P}_1 \cdot \vec{X}(T)) | \vec{X}(t) = \vec{x}] \quad (69)$$

with a control given by:  $\alpha(t, x) = 1_{u_A(t, x) > g(x)}$ .

We note that at  $t = T$  (expiry date),  $\alpha(T, \vec{X}(T)) = \bar{\alpha}(T, \vec{P}_1 \cdot \vec{X}(T)) = 0$  and  $\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) = u_A(T, \vec{X}(T)) = g(T, \vec{P}_1 \cdot \vec{X}(T))$ .

$$\mathbb{E}[\bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) | \vec{X}(0) = \vec{x}_0] = \mathbb{E}[g(\vec{P}_1 \cdot \vec{X}(T)) | \vec{X}(0) = \vec{x}_0] = u_A(0, \vec{x}_0) \quad (70)$$

$$\begin{aligned} u_A(0, x) - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) &= \mathbb{E} \left[ \bar{u}_A(T, \vec{P}_1 \cdot \vec{X}(T)) \mid X(0) = x_0 \right] - \bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) \\ &= \mathbb{E}_{0, x_0} \left[ \int_0^T dt \partial_t \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) + [\mathcal{L} \bar{u}_A](t, x) \right] \\ &= \mathbb{E}_{0, x_0} \left[ \int_0^T dt -\bar{\mathcal{H}}(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))) \right. \\ &\quad \left. + [\mathcal{L} \bar{u}_A](t, x) \right] \\ &\leq \mathbb{E}_{0, x_0} \left[ \int_0^T dt -\bar{\mathcal{H}}(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))) \right. \\ &\quad \left. + \mathcal{H}(t, \vec{X}(t), \vec{P}_1 D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), \vec{P}_1 D^2 \bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1^T) \right] \end{aligned} \quad (71)$$

The operator  $\mathcal{L}v(x, t)$  is given by:

$$\mathcal{L}v(t, x) = \vec{a} \cdot (\vec{P}_1 \partial_x) v(t, x) + \frac{1}{2} Tr[bb^T (\vec{P}_1 \partial_{xx}^2 v(t, x) \vec{P}_1^T)] \quad (72)$$

In the proof we have used that  $\bar{u}_A$  satisfies the HJB equation given by:

$$-\partial_t \bar{u}_A(t, x) = (\bar{\mathcal{H}} \bar{u}_A)(t, x), \quad (t, x) \in [0, T] \times D, \quad (73)$$

$$\bar{u}_A(t, \cdot) = g(\cdot) \quad (74)$$

and that  $[\mathcal{L} \bar{u}_A](t, x) \leq \mathcal{H}$ . Analogously, we may prove the other inequality (67) using that the error may be re-written in the form:

$$\bar{u}_A(0, \vec{P}_1 \cdot \vec{x}_0) - u_A(0, \vec{x}) = \mathbb{E} \left[ u_A(T, \vec{P}_1 \bar{S}(T)) \mid \bar{S}(0) = \vec{P}_1 \cdot \vec{x}_0 \right] - u_A(0, \vec{P}_1 \bar{S}(0)) \quad (75)$$

where we assume that the basket vector  $\vec{P}_1$  is normalised.

### 9.2.2 Source of error in pricing American options using Markovian projection

We now show that if we assume that  $\alpha$  is the optimal control for  $u_A$ , then the projected coefficients must satisfy:

$$\bar{\alpha}^{x_0}(t, s) \bar{\alpha}^{x_0}(t, s) = \mathbb{E}[\alpha(t, \vec{X}(t)) \vec{P}_1 \cdot \vec{a}(t, \vec{x}(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (76)$$

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\alpha(t, \vec{X}(t)) \vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (77)$$

**Proof:** Pricing an American Option involves the optimal control of an SDE using a parameter  $\alpha \in \{0, 1\}$ . If we assume that alpha is fixed and known then we obtain a standard SDE for the underlying process is simply given by:

$$dS_1(t) = \alpha(t, X(t)) \vec{P}_1 \cdot \vec{a}(t, \vec{X}(t)) dt + \alpha(t, \vec{X}(t)) \vec{P}_1^T b(t, \vec{X}(t)) d\vec{W}(t), \quad 0 \leq t \leq T \quad (78)$$

$$\vec{S}_1(0) = \vec{P}_1 \cdot \vec{x}_0 \quad (79)$$

With  $\alpha(t, \vec{X}(t)) \in \{0, 1\}$ . By the Lemma 2 we have that:

$$\bar{\alpha}^{x_0}(t, s) \bar{\alpha}^{x_0}(t, s) = \mathbb{E}[\alpha(t, \vec{X}(t)) \vec{P}_1 \cdot \vec{a}(t, \vec{x}(t)) | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (80)$$

$$(\bar{\alpha}^{x_0}(t, s))^2 (\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\alpha^2(t, \vec{X}(t)) \vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \quad (81)$$

Using that  $\alpha^2(t, \vec{X}(t)) = \alpha(t, \vec{X}(t))$  and  $(\bar{\alpha}^{x_0}(t, s))^2 = \bar{\alpha}^{x_0}(t, s)$  we obtain the required result.

From this we can conclude the Markovian projection error arises because one must project the *combination* of the control  $\alpha(t, \vec{X}(t))$  and  $\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t))$  (and likewise for the diffusion coefficient). This means that, in general, simply projecting only  $\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t))$  and the equivalent diffusion term will not satisfy the above equations and hence not yield the correct coefficients for Markovian projection SDE (correct coefficients are those for which the weak error vanishes). Practically, however,  $\alpha$  being unknown at the outset makes it very difficult to satisfy these equations.

### 9.2.3 Vanishing of error

We want to show that the integrands vanish in regions where the original process and the projected on take the same decision. This entails that Markovian projection errors will arise only in the regions where the control processes make different decisions.

1. Both in stopping region: both Hamiltonians  $\mathcal{H}$  and  $\bar{\mathcal{H}}$  in both integrands will be 0 so the error is zero.
2. One in stopping region and one in continuation region: then we have the error will, in general, be non-zero since one of the Hamiltonians vanishes but not the other.
3. Both in the continuation region: so  $\alpha = \bar{\alpha} = 1$ .

$$\begin{aligned} I_1 &= \int_0^T dt \mathbb{E}_{0, x_0} [\bar{\mathcal{H}}(t, \vec{P}_1 \cdot \vec{X}(t), D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t))) \\ &\quad - \mathcal{H}(t, \vec{X}(t), \vec{P}_1 D\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)), \vec{P}_1 D^2\bar{u}_A(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1^T)] \\ &= \int_0^T dt \mathbb{E}_{0, x_0} [[\mathcal{L}\bar{u}_A](t, x) - [\bar{\mathcal{L}}\bar{u}_A](t, x)] \underbrace{\quad}_{\text{By our choice of } \bar{a} \text{ and } \bar{b}} \quad 0 \end{aligned} \quad (82)$$

An analogous proof holds for the other integral/ inequality.

### 9.2.4 Upper and Lower Bounds on the American Option price

**Lower Bound:** [4] defines two classes of hitting times for the projected value function  $\bar{u}_A$ :

$$\bar{\tau}^* \equiv \inf \left\{ t \in [0, T] : \bar{u}_A \left( t, \bar{S}^{(x_0)}(t) \right) = g \left( \bar{S}^{(x_0)}(t) \right) \right\}, \quad (83)$$

$$\bar{\tau}^\dagger \equiv \inf \left\{ t \in [0, T] : \bar{u}_A \left( t, \bar{P}_1 \cdot \bar{X}(t) \right) = g \left( \bar{P}_1 \cdot \bar{X}(t) \right) \right\}. \quad (84)$$

We also have the hitting time for the value function of the full pricing process  $u_A$  given by:

$$\tau^* \equiv \inf \left\{ t \in [0, T] : u_A \left( t, P_1 \cdot X(t) \right) = g \left( P_1 \cdot X(t) \right) \right\}. \quad (85)$$

and the class of hitting times  $\mathcal{T}_0$ :

$$\mathcal{T}_0 = \{ \tau : \Omega \rightarrow [0, T] \mid \{ \tau \leq t \} \in \mathcal{F}_t \text{ for all } t \in [0, T] \}. \quad (86)$$

$\bar{\tau}^* \in \bar{\mathcal{F}}_t$  but  $\bar{\tau}^\dagger \in \mathcal{F}_t$ . Since  $u_A$  is defined as the stopping time which gives the supremum we see that any  $\tau \in \mathcal{T}_0$  gives a lower bound to the American option price. Since  $\bar{\tau}^\dagger \in \mathcal{T}_0$  we obtain lower bound of the American option price of:

$$u_A(0, x_0) \geq \mathbb{E} \left[ \exp(-r \bar{\tau}^\dagger) g \left( \bar{P}_1 \cdot \bar{X}(\bar{\tau}^\dagger) \right) \mid X(0) = x_0 \right]. \quad (87)$$

**Upper Bound:**

**Theorem 6.** *Rogers's Dual-Representation of the American Option Price [5]: Let  $\bar{Z}(t) = e^{-rt} g(\bar{P}_1 \cdot \bar{X}(t))$ , let  $H_0^1$  denote the space of all integrable martingales  $M$  such that:*

$$\mathbb{E} \left[ \sup_{0 \leq t \leq T} |M(t)| \right] < \infty \text{ and } M(0) = 0 \quad (88)$$

*Then the price of the American option is given by:*

$$u_A(0, \bar{x}_0) = \inf_{M \in H_0^1} \mathbb{E} \left[ \sup_{0 \leq t \leq T} (\bar{Z}(t) - M(t)) \mid \bar{X}(0) = \bar{x}_0 \right] \quad (89)$$

**Proof:**

$$\begin{aligned} Y_0 &= \sup_{0 \leq \tau \leq T} \mathbb{E}[Z_\tau] \\ &= \sup_{0 \leq \tau \leq T} \mathbb{E}[Z_\tau - M_\tau] \\ &\leq \mathbb{E} \left[ \sup_{0 \leq t \leq T} (Z_t - M_t) \right] \end{aligned} \quad (90)$$

Where we have used the since  $M(0) = 0$ ,  $\mathbb{E}[M_\tau] = \mathbb{E}[M_0] = 0$  and for any random variable  $X_t$ ,  $\sup_{\tau} \mathbb{E}[X_\tau] \leq \mathbb{E}[\sup_{\tau} X_\tau]$ . Taking the infimum over all  $M \in H_0^1$  then proves that  $Y_0$  is bounded by  $\mathbb{E} \left[ \sup_{0 \leq t \leq T} (Z_t - M_t) \right]$ .

We use that  $Y_t$  has a Doob-Meyer decomposition:

$$Z_t \leq Y_t = Y_0 + M_t^* - A_t^* \quad (91)$$

where  $M_t^*$  is a martingale vanishing at 0 and  $A_t^*$  is a previsible<sup>1</sup>, integrable increasing process also vanishing at 0.

$$\begin{aligned} \inf_{M \in H_0^1} \mathbb{E} \left[ \sup_{0 \leq t \leq T} (Z_t - M_t) \right] &\leq \mathbb{E} \left[ \sup_{0 \leq t \leq T} (Z_t - M_t^*) \right] \\ &\leq \mathbb{E} \left[ \sup_{0 \leq t \leq T} (Y_t - M_t^*) \right] \\ &\leq \mathbb{E} \left[ \sup_{0 \leq t \leq T} (Y_0 - A_t^*) \right] \\ &= Y_0 \end{aligned} \quad (92)$$

---

<sup>1</sup>A process  $X_t$  adapted to a filtration  $\mathcal{F}_t$  is previsible if, for each time  $t$ ,  $X_t$  is measurable with respect to the  $\sigma$ -algebra generated by all events that happen strictly before  $t$ . You can "predict" the value of the process at time  $t$  using information available strictly before time  $t$ .

Where we have used that since  $A_t^*$  is an increasing process:  $\sup_{0 \leq t \leq T} (Y_0 - A_t^*) = (Y_0 - A_0^*) = Y_0$ . The optimising

martingale (the martingale reaching the infimum) can be written as:

$$dR^*(t) = e^{-rt} ((\nabla u_A)^T b)(t, X(t)) dW(t), \quad 0 < t < T \quad (93)$$

$$R^*(0) = 0. \quad (94)$$

Finding an optimising martingale can be very complex, so, [?] propose to approximate the optimising martingale  $R(t)$  by a near optimal martingale  $R^*(t)$ :

$$u_A(0, \mathbf{x}_0) \leq \mathbb{E} \left[ \sup_{0 \leq t \leq T} (\tilde{Z}(t) - R^*(t)) \mid X(0) = \mathbf{x}_0 \right], \quad (95)$$

$$dR^*(t) = \exp(-rt) \left( (\nabla \bar{u}_A)^T(t, \vec{P}_1 \cdot \vec{X}(t)) \vec{P}_1 \cdot \vec{b}(t, X(t)) \right) dW(t), \quad t \in [0, T], \quad (96)$$

$$R^*(0) = 0.$$

### 9.2.5 Least Squares Monte Carlo for American Options

**General methodology for LSMC:**

1. Use the dynamic programming principle to reduce the American option pricing problem to many sub-problems (solving for the American option price backwards from expiry at  $T$ , first at  $t = T - \Delta t$ ).
2. Sample points in space at the time  $T - \Delta t$ .
3. Define a value function via a suitable parametric family of functions. This step is very important as choosing a family of basis functions that poorly approximates will have large impact on the accuracy of the option price. A lot of intuition and insight into the particular problem is required to choose the right parametric family.
4. Perform a regression to obtain the option price at  $t = T - \Delta t$ .
5. Compare this "holding value" to that of the early exercise price - when the latter exceeds the former we obtain a stopping time.

#### Are there cases where LSMC will be obviously better than Markovian projection (MP)?

If you know the right parametric family to use then it is possible but not obvious that LSMC will be better than MP. There are many difficulties with using LSMC including the error propagation

1. The error from the regression estimate of the continuation value is propagated at each step (backwards). The number of samples you need to alleviate the error propagation further grows exponentially and likewise the number of basis functions required also becomes large.
2. Markovian projection is a non-convergent method thus giving us limitation on the error that Markovian projection involves.

## 9.3 Essential Dimensionality:

In this section, we give important results derived in Let  $1 \leq n < d$  and  $D \subset \mathbb{R}^d$  be a convex set with piecewise smooth boundary.

We call a function  $v : D \rightarrow \mathbb{R}$  *essentially  $n$ -dimensional* if there exist a function  $\zeta : \mathbb{R}^n \rightarrow \mathbb{R}$  and a matrix  $\mathbf{N} \in \mathbb{R}^{n \times d}$  with orthogonal rows such that  $v : \mathbb{R}^d \rightarrow \mathbb{R}$  is given by

$$v(x) = \zeta(\mathbf{N}x).$$

By extension, we call a differential operator  $\mathcal{K}$  essentially  $n$ -dimensional if the following backward PDE is well posed:

$$-\partial_t w(t, x) = \mathcal{K}w(t, x), \quad (t, x) \in [0, T) \times D, \quad w(T, \cdot) = w_T(\cdot), \quad (284)$$

and it has a unique essentially  $n$ -dimensional solution for any essentially  $n$ -dimensional terminal value  $w_T$ .

Here we specifically mean that the function  $\zeta$  may depend on time, that is,

$$w(t, x) = \zeta(t, \mathbf{N}x),$$

but the matrix  $\mathbf{N}$  does not.

**Lemma 3. Separating essential dimensionality and early exercise** If a model has a pricing PDE whose differential operator  $\mathcal{L}$  is essentially  $n$ -dimensional then an American option price is also essentially  $n$ -dimensional if the corresponding payoff is essentially  $n$ -dimensional in the same subspace.

**Proof:** We begin by approximating the price of an American option by a Bermudan option that can only be exercised at  $M$  equally-spaced intervals:  $t_m = m \frac{T}{M}$ . Assume that our model has a pricing PDE whose differential operator  $\mathcal{L}$  is essentially  $n$ -dimensional and that our payoff function  $g = g(Px)$ .

Base case: consider time interval  $t_{M-1} < t < T$ . In this time interval, the option is European. By assumption, our option price in this time interval therefore is essentially  $n$ -dimensional so may be expressed as:

$$u(t, x) = \bar{u}(t, Px), \quad t_{M-1} < t < T \quad (97)$$

So the value of our option at exercise time  $t_{M-1}$  is the maximum of two essentially one-dimensional function:

$$u(t_{M-1}) = \max(u(t_{M-1}^+, x), g(Px)) = \max(\bar{u}(t_{M-1}^+, Px), g(Px)) \quad (98)$$

Now consider an exercise time  $t_m$  where we assume that the option price in the interval  $t_m < t < t_{m+1}$  is essentially  $n$ -dimensional. Then the price of the Bermudan option at a time  $t_m$  is given by:

$$u(t_m^-) = \max(u(t_m^+, x), g(Px)) = \max(\bar{u}(t_m^+, Px), g(Px)) = \bar{u}(t_m, Px) \quad (99)$$

From this we can conclude that the price of the Bermudan at all times will be essentially  $n$ -dimensional, assuming that its pricing PDE and the payoff are both essentially  $n$ -dimensional on the same subspace.

The American option price is the limit of the Bermudan option price as  $M \rightarrow \infty$ . Hence, we conclude that given our assumption, the American option price is also essentially  $n$ -dimensional.

### 9.3.1 Relationship between dimensional reduction and exactness of Markovian projection for American Option pricing

**Lemma 4.** If a model has a pricing PDE whose coefficients are essentially  $n$ -dimensional and the American option price is also essentially  $n$ -dimensional on the same sub-space, then the Markovian projection on that  $n$ -dimensional subspace is exact, namely it produces exact values of the American option price.

**Proof:** First we prove that the Hamiltonians corresponding to  $u_A$  and  $\bar{u}_A$  coincide pointwise for the case  $n = 1$  but this proof may be easily extended to higher  $n$ , that is:

$$\mathcal{H}(t, s\vec{P}, v\vec{P}, \vec{P}Q\vec{P}^T) - \bar{\mathcal{H}}(t, s, v, Q) = 0 \quad (100)$$

Where we recall that  $\mathcal{H}$ :

$$\mathcal{H} = \max_{\alpha \in \{0,1\}} (\alpha \mathcal{L}u_A(t, \vec{x})) \quad (101)$$

$$\mathcal{L}v(t, \vec{x}) = (\vec{a}_i \partial_{x_i})v(t, \vec{x}) + \frac{1}{2}(bb^T)_{ij}(\partial_{x_i x_j}^2 v(t, \vec{x})) \quad (102)$$

Assume that the model's pricing PDE has coefficients which are essentially 1-dimensional then:

$$\begin{aligned}\bar{a}(t, \vec{x}) &= \bar{a}'(t, \vec{P} \cdot \vec{x}) \\ \bar{b}(t, \vec{x}) &= \bar{b}'(t, \vec{P} \cdot \vec{x})\end{aligned}\tag{103}$$

Assume also that the American option price is essentially 1-dimensional such that  $u_A(t, \vec{x}) = \bar{u}_A(t, \vec{P} \cdot \vec{x})$ . We now consider the coefficients of the Markovian projection:

$$\begin{aligned}\bar{a}^{(x_0)}(t, s) &= \mathbb{E}[\vec{P} \cdot \bar{a}(t, \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[\vec{P} \cdot \bar{a}'(t, \vec{P} \cdot \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \vec{P} \cdot \bar{a}'(t, s)\end{aligned}\tag{104}$$

$$\begin{aligned}(\bar{b}^{(x_0)}(t, s))^2 &= \mathbb{E}[(\vec{P}^T b b^T \vec{P})(t, \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[(\vec{P}^T b' b'^T \vec{P})(t, \vec{P} \cdot \vec{X}(t)) \mid \vec{P} \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= (\vec{P}^T b' b'^T \vec{P})(t, s)\end{aligned}\tag{105}$$

From this we can see that the SDE for the true process  $S = \vec{P} \cdot \vec{X}$  and the SDE for the projected process  $\bar{S}$  will coincide exactly. We can now compare the Hamiltonians:

$$\begin{aligned}\mathcal{H}(t, s, \vec{P}, v, \vec{P}, \vec{P} Q \vec{P}^T) &= \left( \bar{a}'(t, s) \cdot \vec{P} v + \frac{1}{2} \text{Tr}[(b' b'^T \vec{P} Q \vec{P}^T)] \right)^+, \\ \bar{\mathcal{H}}(t, s, v, H) &= \left( \bar{a}(t, s) v + \frac{1}{2} \bar{b}^2(t, s) Q \right)^+, \end{aligned}\tag{106}$$

Using the expressions for our coefficients derived above we see that these Hamiltonian's coincide pointwise. Since the two Hamiltonians coincide *pointwise* not just in expectation, the error bounds derived previously imply that:

$$u_A(t, \vec{x}) = \bar{u}_A(t, \vec{P} \cdot \vec{x}), \quad \forall (t, x) \in [0, T] \times D.$$

Hence, we conclude that if the American option price is essentially n-dimensional then the Markovian projection on that n-dimensional subspace is exact, i.e. it produces exact values of the American option price.

## 9.4 Bachelier Model:

### 9.4.1 Exactness of Markovian Projection for pricing American Options under the Bachelier Model:

Risk-free measure and no arbitrage give:

$$\bar{a} = r \bar{s}\tag{107}$$

Bachelier Model chooses:

$$b_{ij} = \Sigma_{ij}\tag{108}$$

with  $\Sigma$  a constant matrix. This yields that  $\vec{X}(t)$  is Gaussian distributed with  $\vec{X}(t) \sim \mathcal{N}(\vec{\mu}, \Gamma(t))$ . The projection  $S(t) = \vec{P} \cdot \vec{X}(t)$  is again Gaussian distributed with  $S(t) \sim \mathcal{N}(\vec{P} \cdot \vec{\mu}, \vec{P}^T \Gamma(t) \vec{P})$ . We can then use the following important property of Gaussian random variables that :

**Proposition.** If random variables  $Y$  and  $Z$  follow a *bivariate Gaussian* distribution with means  $\mu_Y, \mu_Z$ , standard deviations  $\sigma_Y, \sigma_Z$ , and correlation  $\rho$ , then

$$\mathbb{E}[Y \mid Z = z] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_Z} (z - \mu_Z).$$

Thus, the conditional expectation of a Gaussian random variable is *affine* in the conditioning variable  $Z$ .



If we take  $Z = S(t)$  and  $Y = S(t+h)$  then we can show that the conditional mean and variance depend only on  $Z$ , yielding that:

$$\mathbb{P}(S_1(t+h) \in A \mid \mathcal{F}_t) = \mathbb{P}(S_1(t+h) \in A \mid S_1(t)), \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

This means that the process  $S(t)$  is Markovian. Hence, the Markovian projection of the Bachelier Model will give exact results for the true process  $S(t)$ : the Markovian projection yields an SDE which coincides with the underlying high-dimensional portfolio in law.

#### 9.4.2 Evaluation of coefficients:

The coefficient  $\vec{a}(t, \vec{X}(t)) = r\vec{X}(t)$

$$\begin{aligned} \bar{a}^{x_0}(t, s) &= \mathbb{E}[\vec{P}_1 \cdot \vec{a}(t, \vec{X}(t)) \mid \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= \mathbb{E}[r\vec{P}_1 \cdot \vec{X}(t) \mid \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &= rs \end{aligned} \tag{109}$$

An additional consequence of having a constant volatility matrix is that the coefficient  $\bar{b}$  of the projected process can be evaluated without needing the Laplace approximation to compute high-dimensional integrals:

$$\begin{aligned} (\bar{b}^{x_0}(t, s))^2 &= \mathbb{E}[\vec{P}_1^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P}_1 \mid \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \\ &\underset{\substack{= \\ \text{Bachelier}}}{=} \vec{P}_1^T \Sigma \Sigma^T \vec{P}_1 \end{aligned} \tag{110}$$

## 10 Exercise Rate Optimisation:

In this section, we follow [6] in giving a brief overview of the current state of the art methods to determine exercise regions for American options. We then give the key results derived in the paper.

$$u(0, s_0) = \sup_{E \subset \mathbb{R}^{1+d}} \mathbb{E}[Y_{\tau_E T} \mid S_0 = s_0] \tag{111}$$

pick a set  $E$  in the time-state plane; stop the first time your path  $(t, S_t)$  hits that set; compute the expected payoff; then choose the set giving the largest expectation.

Markov property implies an optimal stopping rule can be chosen to depend only on the current time and current state, not on the full path history. Therefore, any optimal stopping time can be represented as the first entry time into a stopping region  $E$ :

$$E = \{(t, s) : v(t, s) = Y_t(s)\} \tag{112}$$

General strategy to find an option price:

1. Choose a parametrisation of the subsets of  $\mathbb{R}^{1+d}$
2. Choose an initial guess  $E_0 \subset \mathbb{R}^{1+d}$
3. Update to get  $E_n \rightarrow E_\infty$  and  $\Psi(E_n) \rightarrow \Psi(E_\infty) = u(s_0)$

Problems with this general strategy:

1. Difficult to come up with a parametrisation all possible exercise regions for path-dependent (exotic) or multi-dimensional options. Analytic guesses possible for simple options but in higher dimensions this is difficult because exercise regions may be disjoint.

2. Unclear to determine the global optimum, given a parametrisation.
3. In general, expectations are replaced with empirical averages. Finite sample paths means that \*\*\* is highly noisy and irregular. Sampling noise - when you scan many candidate sets  $E$  each  $\hat{\psi}(E)$ , the largest observed  $\hat{\psi}(E)$  will be the one where the noise happened to push the sample average up - selection maximisation bias.

Aim of the paper is to relax the optimisation problem by replacing exercise regions with exercise rates, defining randomised exercise strategies.

## 10.1 Longstaff-Schwartz Algorithm:

- sensitive to the number of exercise dates (backwards iteration in time):

Let the exercise dates be  $t_0 < t_1 < \dots < t_N = T$ . Suppose we have an approximation  $v_n(\cdot)$  to the value function  $v(t_n, \cdot)$  that can be evaluated at arbitrary state points. Proceed backward in  $n = N, N-1, \dots, 1$  as follows (showing one generic step at  $t_{n-1}$ ):

1. Generate  $M$  samples  $s^{(m)} \sim S_{t_{n-1}}$ ,  $m = 1, \dots, M$ .
2. For each  $s^{(m)}$ , generate  $K$  conditional forward samples  $s^{(m,k)} \sim S_{t_n} \mid S_{t_{n-1}} = s^{(m)}$ ,  $k = 1, \dots, K$ .
3. Estimate the *continuation value* from  $s^{(m)}$  by the nested Monte Carlo average

$$c^{(m)} := \frac{1}{K} \sum_{k=1}^K v_n(s^{(m,k)}).$$

4. Regress (discrete least squares) onto a chosen approximation space  $V$  (e.g. polynomials):

$$p_{n-1} := \arg \min_{p \in V} \sum_{m=1}^M |p(s^{(m)}) - c^{(m)}|^2.$$

Like in the  $\mathcal{L}^2$  regression exercise, you choose a set of basis function  $\phi_{i_1, i_2}(t, x)$  and fit coefficients  $\beta_{i_1, i_2}$  and then do least-squares regression to minimise the error between the expected (discounted) continuation.

5. Define the step-back value function by the dynamic programming relation

$$v_{n-1}(s) := \max\{g(s), p_{n-1}(s)\}.$$

For each path you compare the fitted continuation value and immediate exercise payoff.

Iterate backward to obtain  $v_0(\cdot)$ , an estimate of the option value at time 0.

Main issues with the Longstaff-Schwartz algorithm:

1. Value function has only one continuous derivative at the boundary  $E_\infty$ . Polynomial regression is poor at (low accuracy in) approximating functions with kinks/low regularity unless the basis is rich (large ansatz space) and many samples necessary.
2. The error from the regression estimate of the continuation value is propagated at each step (backwards). The number of samples you need to alleviate the error propagation further grows exponentially.

## 10.2 Exercise Rate Optimisation:

# 11 Multi-Level Monte Carlo (MLMC)

## 11.1 Introduction:

In this section, we give an overview of the key results on Multi-Level Monte Carlo from [7] and [8]. When do we use Monte Carlo methods?

1. Uncertainty is of high dimension - which can also mean that the number of uncertain input parameters is high.
2. Uncertainty is strongly non-linear.

### 11.1.1 Standard (simple) Monte Carlo:

Under standard Monte Carlo the estimator of  $\mathbb{E}[P]$  is just an average of values of  $P(\omega)$  for  $N$  independent samples from the probability space  $(\Omega, \mathcal{F}, P)$  given by:

$$\mathbb{E}[\hat{P}] = N^{-1} \sum_{i=1}^N P(\omega^{(n)}) \quad (113)$$

The variance of the estimate is given by  $N^{-1}\mathbb{V}[P]$ , giving an rms error of  $\mathcal{O}(N^{-\frac{1}{2}})$ . An accuracy of  $\epsilon$  therefore needs  $N = \mathcal{O}(\epsilon^2)$  samples.

### 11.1.2 Control Variates:

### 11.1.3 General Idea for Multi-Level Monte Carlo:

Suppose we have a sequence  $P_0, \dots, P_{L-1}$  which approximates  $P_L$  with increasing accuracy and cost. Then we can write the expected value of  $P_L$  using a telescoping sum:

$$\mathbb{E}[P_L] = \mathbb{E}[P_0] + \sum_{l=1}^L \mathbb{E}[P_l - P_{l-1}] \quad (114)$$

This means that we can use the following unbiased estimator of  $P_L$ :

$$Y = N_0^{-1} \sum_{n=1}^{N_0} P_0^{(0,n)} + \sum_{\ell=1}^L \left\{ N_\ell^{-1} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}) \right\}. \quad (115)$$

The total cost and the total variance are then given by (assuming independent samples):

$$\text{Cost} = \sum_{\ell=0}^L N_\ell C_\ell, \quad \mathbb{V}(Y) = \sum_{\ell=0}^L N_\ell^{-1} V_\ell.$$

Assuming a fixed variance, the cost is minimised by choosing  $N_\ell$  to minimise:

$$\sum_{\ell=0}^L (N_\ell C_\ell + \mu^2 N_\ell^{-1} V_\ell),$$

with  $\mu^2$  a Lagrange multiplier. This gives:

$$N_\ell = \mu \sqrt{\frac{V_\ell}{C_\ell}},$$

Assuming we want an overall variance given by  $\epsilon^2$  we require:

$$\mu = \epsilon^{-2} \sum_{\ell=0}^L \sqrt{V_\ell C_\ell},$$

Hence, the optimal total computational cost is given by:

$$C_{\text{total}} = \epsilon^{-2} \left( \sum_{\ell=0}^L \sqrt{V_\ell C_\ell} \right)^2. \quad (1.1)$$

What is determined by the particular problem is the relative rate at which the variance decreases with level compared to the increase in the computational cost.

#### 11.1.4 Geometric MLMC:

When the sample space is infinite, such as when we want to simulate SDEs, the output on the level  $P_\ell$  on level  $\ell$  is an approximation of the random variable  $P$ , which cannot be simulated.

Assuming that  $Y$  is an approximation to  $\mathbb{E}[P]$ , then the Mean Square Error (MSE) is given by:

$$\text{MSE} \equiv \mathbb{E}[(Y - \mathbb{E}[P])^2] = \mathbb{V}[Y] + (\mathbb{E}[Y] - \mathbb{E}[P])^2.$$

where the first contribution to the error is the statistical sampling error which measures the variance from using a finite number of Monte Carlo samples and the second is the weak error (or bias), which arises from stopping the telescoping sum at level  $L$ . We then assume that  $Y$  is the multi-level estimator given by:

$$Y = \sum_{\ell=0}^L Y_\ell, \quad Y_\ell = \frac{1}{N_\ell} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}), \quad P_{-1} \equiv 0.$$

then we obtain that:

$$\mathbb{E}[Y] = \mathbb{E}[P_L], \quad \mathbb{V}[Y] = \sum_{\ell=0}^L \frac{V_\ell}{N_\ell}, \quad V_\ell \equiv \mathbb{V}[P_\ell - P_{\ell-1}].$$

To ensure that the MSE is less than  $\epsilon^2$ , it is sufficient to ensure that each contribution to the MSE is less than  $\epsilon^2$ . [7] then assumes a geometric sequence of levels with the cost increasing exponentially with levels and variance decreasing exponentially with levels.

#### Giles' MLMC Theorem

**Theorem 7.** Let  $P$  be a random variable and  $P_\ell$  its level- $\ell$  approximation. Suppose there exist independent estimators  $Y_\ell$  based on  $N_\ell$  samples, each of cost  $C_\ell$  and variance  $V_\ell$ , and positive constants  $\alpha, \beta, \gamma, c_1, c_2, c_3$  with  $\alpha \geq \frac{1}{2} \min(\beta, \gamma)$  such that

$$\textbf{Weak Error: } |\mathbb{E}[P_\ell - P]| \leq c_1 2^{-\alpha\ell},$$

$$\mathbb{E}[Y_\ell] = \begin{cases} \mathbb{E}[P_0], & \ell = 0, \\ \mathbb{E}[P_\ell - P_{\ell-1}], & \ell > 0, \end{cases}$$

$$\textbf{Variance: } V_\ell \leq c_2 2^{-\beta\ell},$$

$$\textbf{Cost: } C_\ell \leq c_3 2^{\gamma\ell}.$$

Then for any  $\varepsilon < e^{-1}$  there exist  $L$  and  $\{N_\ell\}$  such that the estimator

$$Y = \sum_{\ell=0}^L Y_\ell$$

satisfies

$$\mathbb{E}[(Y - \mathbb{E}[P])^2] < \varepsilon^2,$$

and its cost  $C$  obeys

$$\mathbb{E}[C] \leq \begin{cases} c_4 \varepsilon^{-2}, & \beta > \gamma, \\ c_4 \varepsilon^{-2} (\log \varepsilon)^2, & \beta = \gamma, \\ c_4 \varepsilon^{-2 - (\gamma - \beta)/\alpha}, & \beta < \gamma. \end{cases}$$

#### Sketch of the proof:

Each of the constraints on the size of the weak error and the statistical error gives a constraint on the number of levels  $L$  and the number of samples per level  $N_\ell$  respectively.

1.  $\mathbb{E}[Y - P] = \mathbb{E}[\sum_{\ell} Y_\ell - P] = \mathbb{E}[P_L - P] \leq c_1 2^{-2\alpha L} < \frac{1}{\sqrt{2}}\varepsilon$ . This gives that  $L \gtrsim \frac{1}{\alpha} \log_2(\frac{1}{\varepsilon})$ .
2. We obtain that the optimal  $N_\ell$  is given by:  $N_\ell = \frac{2}{\varepsilon^2} (\sum_{\ell'=0}^L \sqrt{V_{\ell'} C_{\ell'}}) \sqrt{\frac{V_\ell}{C_\ell}}$ .

The optimal computational cost is then, assuming that  $N_\ell \propto 2^{-(\beta+\gamma)\frac{\ell}{2}}$  and  $N_\ell C_\ell \propto 2^{(-\beta+\gamma)\frac{\ell}{2}}$ :

$$C \lesssim \frac{1}{\varepsilon^2} \left( \sum_{\ell'=0}^L \sqrt{V_{\ell'} C_{\ell'}} \right)^2 \quad (116)$$

We now have three cases depending on the relative magnitudes of  $\beta$  and  $\gamma$ :

1.  $\beta > \gamma$ : then we have a sequence that converges so the brackets are constant. Hence  $C \lesssim c_4 \varepsilon^{-2}$ . In this case, most of the computational cost is from the coarse levels where  $\ell$  is low. To further improve the computational cost we can note that adding the further levels will not significantly add to the computational cost. Hence, the equal error split is sub-optimal. Instead, we may give more error to the statistical error/variance term and so fewer samples  $N_\ell$  will be required.
2.  $\beta = \gamma$ : then we have  $C \lesssim k \varepsilon^{-2} (L+1)^2 \approx c_4 \varepsilon^{-2} \log(\varepsilon)^2$ . In this case, computational cost and contribution to the variance is evenly spread across all the levels.
3.  $\beta < \gamma$ : then the dominant contribution to the sum is from level  $L$ . So we obtain:  $C \lesssim k \varepsilon^{-2} 2^{-(\beta-\gamma)L} \approx \varepsilon^{-2 - \frac{(-\beta+\gamma)}{\alpha}}$ . In this case the dominant computational cost is from the finest levels.

We note that the computational cost for standard Monte Carlo is  $\mathcal{O}(\varepsilon^{-2 - \frac{\gamma}{\alpha}})$ .

#### 11.1.5 Multi-Index Monte Carlo (MIMC):

### 11.2 Multi-Level Monte Carlo for SDEs:

We consider SDE's of the form:

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t \\ S_0 &= s_0 \end{aligned} \quad (117)$$

We then want to compute the expected value of  $f(S(T))$ , where the function  $f$  is assumed for now to be Lipschitz.

A simple Euler-Maruyama discretisation with time step  $h$  of the SDE gives:

$$\hat{S}_{n+1} = \hat{S}_n + a(\hat{S}_n, t_n) h + b(\hat{S}_n, t_n) \Delta W_n.$$

### 11.2.1 Standard Monte Carlo:

The standard Monte Carlo estimator for  $\mathbb{E}[f(S(T))]$  is given by:

$$\hat{Y} = \frac{1}{N} \sum_{i=1}^N f(\hat{S}_T^{(i)}).$$

where we implicitly assume a certain step-size  $h$ . Assuming that  $a(S_t, t)$  and  $b(S_t, t)$  satisfy certain conditions (Bally and Talay 1995, Kloeden and Platen 1992, Talay and Tubaro 1990) then the MSE is given by:

$$MSE \approx c_1 N^{-1} + c_2 h^2 \quad (118)$$

The first term corresponds to the statistical sampling error (variance in the estimator  $\hat{Y}$ ) and the second term the weak error due to error induced by the Euler time discretisation which is  $\mathcal{O}(h)$ . Setting  $MSE = \epsilon^2$ , we require  $N = \mathcal{O}(\epsilon^{-2})$  and  $h = \mathcal{O}(\epsilon)$ . This gives a total computational cost  $C_{MC} = \mathcal{O}(\epsilon^{-3})$

### 11.2.2 Multi-Level Monte Carlo for Euler-discretised SDE:

Giles' MLMC Method:

1. We take a geometric sequence of levels such that at level  $\ell$  the uniform time-step is taken to be  $h_\ell = h_0 M^{-\ell}$  with  $M$  a positive integer typically taken to be 2.  $h_0$  may be taken to be the entire time interval  $T$  as long as it remains a good enough control variate for the other finer levels.
2. To couple the coarse and the fine levels we use the same underlying Brownian path and sum the Brownian increments for the fine path time steps to obtain the Brownian increments for the coarse time-steps.
3. We can then use that given a Lipschitz payoff function  $f$  for which for any  $S_1, S_2$  there exists a constant  $K$  such that  $|f(S_1) - f(S_2)| \leq K||S_1 - S_2||$  and that the strong error of the Euler discretisation is  $\mathbb{E}[||S - \hat{S}||^2] = \mathcal{O}(h)$  that:

$$\begin{aligned} V_\ell &\equiv \mathbb{V}[P_\ell - P_{\ell-1}] = \mathbb{V}[(P_\ell - P) - (P_{\ell-1} - P)] \\ &\leq 2(\mathbb{V}[P - P_\ell] + \mathbb{V}[P - P_{\ell-1}]) \\ &\leq 2(K^2 \mathbb{E}[||S - \hat{S}_\ell||^2] + K'^2 \mathbb{E}[||S - \hat{S}_{\ell-1}||^2]) \end{aligned} \quad (119)$$

This gives that  $V_\ell = \mathcal{O}(h_\ell)$ . Hence, if we take  $M = 2$  we have  $\alpha = 1, \beta = 1, \gamma = 1$ . This means the complexity to reach a rms error of  $\epsilon^2$  is, by Theorem [7],  $C_{MLMC} = \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$ .

This means the MLMC method is near the optimal lower bound of  $\mathcal{O}(\epsilon^2)$  for complexity, save logarithmic factors.

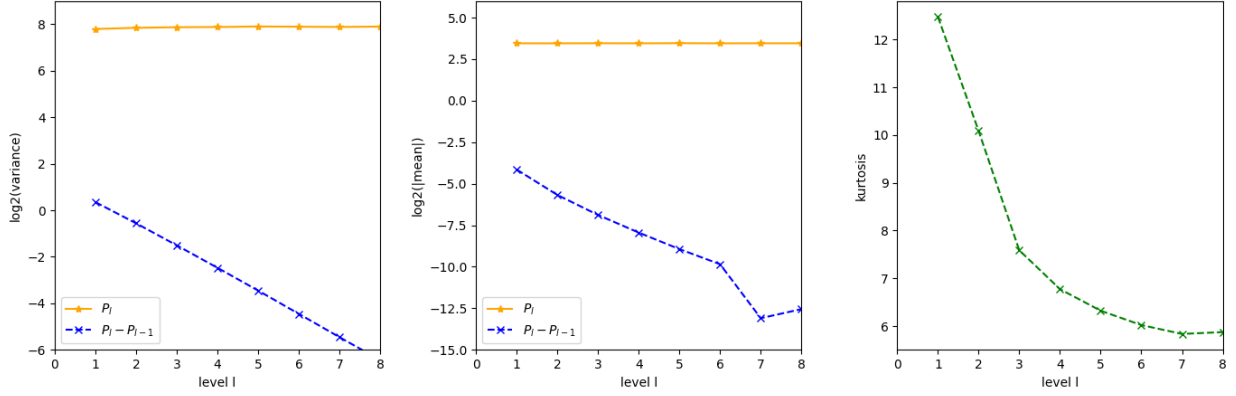
### 11.2.3 Numerical Example: Pricing a European Call Option under GBM

We now price a European Call Option on a single stock, giving pay-off  $f(S_T) = e^{-rT}(S_T - K)^+$  whose underlying asset satisfies Geometric Brownian Motion (GBM) following the methodology outlined in [7].

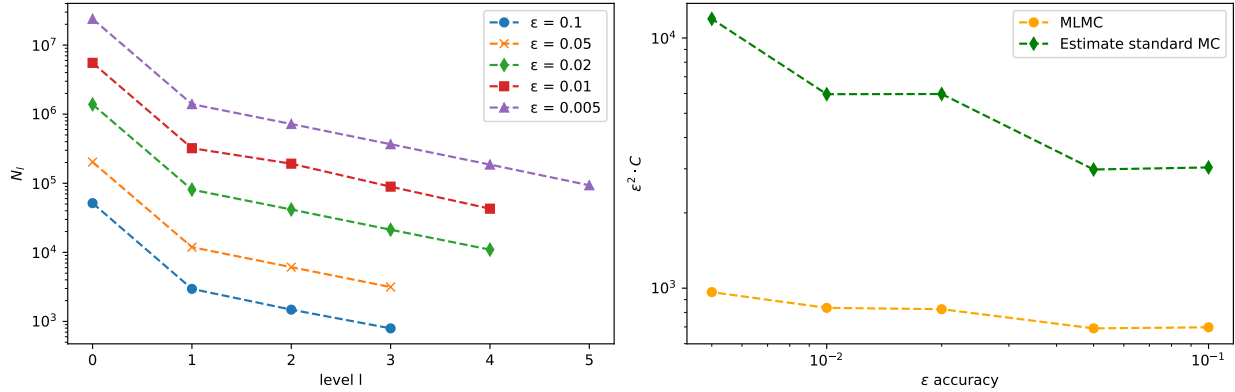
$$dS_t = rS_t + \sigma S_t dW_t \quad (120)$$

Following [7], we choose  $r = 0.05, \sigma = 0.2, S_0 = 100, T = 1, K = 100$ , allowing us to verify that we do in fact correctly implement MLMC.

This contrasts with the standard Monte Carlo cost which is approximately  $\epsilon^{-2} C_L V_0$ .



**Figure 2:** We estimate the variance, absolute mean and kurtosis with level  $\ell$  of an initial sample of  $10^5$  paths. [7,8] checks the kurtosis given by  $\kappa = \frac{\mathbb{E}[X^4]}{(\mathbb{E}[X^2])^2}$  for a random variable  $X$  with zero mean. For a large number of sample, the standard deviation of the sample variance of the random variable  $X$  is  $\sigma_{samplevar} = \sqrt{\frac{\kappa-1}{N} \mathbb{E}[X^2]}$ . It is important to check that  $\kappa$  is small to ensure that a reliable estimate of the variance is obtained.



**Figure 3:** Left hand figure plots number of paths per level for a selected accuracy  $\epsilon$ . Right hand figure plots  $\epsilon^2 * C$  for MLMC and an estimated  $\epsilon^2 * C$  of standard Monte Carlo. Given that the theoretical expected cost is for MLMC for our case is  $C \sim \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$  we expect that  $\epsilon^2 * C$  should be approximately constant as we see in the plot.

#### 11.2.4 Discontinuous Payoffs:

### 11.3 Multi-Level Weighted Least Squares Approximation [1] [2]

#### 11.3.1 Polynomial (Weighted) Least Squares Approximation:

The goal of polynomial weighted least squares (PWLS) is to approximate a function  $f$ ,  $f \in \mathcal{L}_\mu^2(\Gamma)$ , where  $\mu$  is a probability measure on parameter space  $\Gamma$ , using out finitely many point evaluations of  $f$ . The general methodology goes as follows:

1. Choose a finite-dimensional space of polynomials  $V \in \mathcal{L}_\mu^2(\Gamma)$ .
2. Compute  $f(\vec{y}_i)$  for random samples  $(\vec{y}_i)_{i=1}^N$  drawn i.i.d from some distribution  $\nu$  on  $\Gamma$
3. Minimise the discrete least squares error:

$$\begin{aligned} \Pi_V f &:= \arg \min_{v \in V} \|f - v\|_{disc}^2 \\ \|f - v\|_{disc}^2 &:= \frac{1}{N} \sum_{j=1}^N \frac{d\mu}{d\nu} |f(\vec{y}_j) - v(\vec{y}_j)|^2 := \langle f - v, f - v \rangle_{disc} \end{aligned} \quad (121)$$

where the  $\frac{d\mu}{d\nu} := w$  is the weight function and corrects for the difference between the true distribution  $\mu$  and the sampling distribution  $\nu$ .

4. The coefficients  $\vec{c}$  of  $\Pi_V f$  with respect to any basis  $(B_j)_{j=1}^m$  of  $V$  are given by the normal equation:

$$\hat{G} \vec{c} = \vec{v} \quad (122)$$

with  $\hat{G}_{ij} = \langle B_i, B_j \rangle_{disc}$  and  $v_j = \langle f, B_j \rangle_{disc}$  with  $i, j \in \{1, \dots, m\}$  assuming that  $\hat{G}$  is invertible. If  $\hat{G}$  is not invertible, then we define  $\Pi_V f$  as the one with minimal  $\mathcal{L}_\mu^2(\Gamma)$  norm.

**Uniqueness of minimisation** requires that  $N \geq \dim(V) = m$ . If  $\nu = \mu$ , then  $N = \mathcal{O}(m^b \log(m))$  ensures a quasi-optimal polynomial fit, where the specific value of  $b \geq 1$  depends on  $\mu$ . For a specific optimal distribution we can obtain the same result with  $N = Cm \log(m)$  (Hampton and Doonstan (2015)).

#### 11.3.2 Sampling from the optimal distribution:

We want to sample from the optimal distribution  $\nu_V^*$  for which  $N = \mathcal{O}(m \log(m))$ . This occurs for the weight  $w_V^* = \rho_V^{*-1}$  which is associated with density:

$$\rho_V^*(y) := \frac{1}{m} \sum_{j=1}^m |B_j(y)|^2.$$

[1] show that the density of the optimal distribution  $\nu_V^*$  associated with any downward closed<sup>2</sup> polynomial subspace on  $[0, 1]^d$  with respect to the Lebesgue measure  $d\lambda$  satisfies:

$$C^{-d} < \frac{d\nu_V^*}{d\lambda} \leq C^d p_d^\infty,$$

with  $0 < C < \infty$  and  $p_d^\infty(y)$ , the Lebesgue density of the d-dimensional arcsine distribution given by:

$$p_d^\infty(y) := \prod_{j=1}^d \frac{1}{\pi \sqrt{y_j (1 - y_j)}}.$$

---

<sup>2</sup>A subspace is downward closed if you have monomial in your subspace and there is another monomial whose degree is smaller in each component, then this monomial is also in your subspace.



The upper bound implies that the number samples for a near-optimal approximation of  $f$ , assuming the weight function associated with the arcsine distribution, is given by  $C^d m \log(m)$ , where  $C$  is independent of  $V$  (but depends exponentially on  $d$ ).  $C^d$  has been tested to be small even in high dimensions, ( $\approx 4$ ) for  $d = 10$ . **As a consequence the upper bound allows for efficient sampling in many cases (without the need for MCMC methods or acceptance/rejection sampling) from the arcsine distribution rather than the optimal distribution for all polynomial subspaces. It also gives a simple weight function independent of  $V$ , which is important for adaptive algorithms.**

One way to sample from the univariate arcsine distribution:

1. Generate a uniformly distributed random variable  $X \sim U[-\frac{\pi}{2}, \frac{\pi}{2}]$ .
2. Transform  $X$  via:  $Y = \frac{1}{2}(1 + \sin(X))$  then  $Y$  samples the univariate arcsine distribution.

### 11.3.3 Single-Level Approach to Inexact Solutions:

The single-level approach applies the weighted least squares approximation to a single  $f_n$  approximating some function  $f$ . The problem with this approach is that a good approximation requires a large subspace  $V$  and fine time steps leading to many, expensive samples.

It may be shown that if the following two assumptions hold:

1. There are  $\beta, \gamma > 0$  such that  $\|f - f_h\|_{\mathcal{L}_\mu^2(\Gamma)} \lesssim h^\beta$  and each evaluation of  $f_h$  requires work  $h^{-\gamma}, \forall h > 0$ .
2. There is an  $\alpha > 0$  and a sequence of subspaces  $(V_n)_{n=1}^\infty$  such that  $\inf_{v \in V_n} \|f_h - v\|_{\mathcal{L}^\infty(\Gamma)} \lesssim (\dim(V_n))^{-\alpha}$  for all  $h > 0, n \in \mathbb{N}$

then an accuracy of  $\epsilon > 0$  requires  $h \approx \epsilon^{-\frac{\gamma}{\beta}}$  and  $\dim(V_n) \approx \epsilon^{-\frac{1}{\alpha}}$ . The total work therefore is given by:

$$\begin{aligned} &\approx h^{-\gamma} \dim(V_n) \log(\dim(V_n)) \\ &\approx \epsilon^{-\frac{\gamma}{\beta}} \epsilon^{-\frac{1}{\alpha}} |\log(\epsilon)| \end{aligned} \tag{123}$$

Where  $\epsilon^{-\frac{\gamma}{\beta}}$  is the contribution to the work from PDE solutions and  $\epsilon^{-\frac{1}{\alpha}}$  the contribution from the complexity of polynomial spaces.

## 11.4 Multi-Level Approach to Inexact Solutions

### 11.4.1 Core Assumptions:

[1,2] show how the Multi-Level methodology can be applied to improve the computational work required for a weighted least squares approximation.

Authors consider the approximation of a function  $f_\infty : \Gamma \subset \mathbb{R}^d \rightarrow \mathbb{R}, d \in \mathbb{N} \cup \{\infty\}$  in a normed vector space  $(F, \|\cdot\|_F) \hookrightarrow (L_\mu^2(\Gamma), \|\cdot\|_{L_\mu^2(\Gamma)})$  of continuous functions on  $\Gamma$ , under the following assumptions:

**A1** (Convergence of approximations). There exist functions  $f_n \in F, n \geq 1$  such that

$$\begin{aligned} \|f_\infty - f_n\|_F &\lesssim n^{-\beta_s} \\ \|f_\infty - f_n\|_{L_\mu^2(\Gamma)} &\lesssim n^{-\beta_w} \end{aligned}$$

for some  $\beta_s > 0$  and  $\beta_w \geq \beta_s$ .

**A2(p)** (Polynomial approximability). There exist downward closed spaces of polynomials  $V_m, m \geq 1$  on  $\Gamma$  such that

$$\dim V_m \lesssim m^\sigma, \quad e_{m,p}(F) \lesssim m^{-\alpha}$$

for some  $\sigma > 0$ ,  $\alpha > 0$ , and  $p = 2$  or  $p = \infty$ , where

$$e_{m,2}(F) := \sup_{f \in F} \frac{e_{V_m, \|\cdot\|_{L^2_\mu(\Gamma)}}(f)}{\|f\|_F}, \quad e_{m,\infty}(F) := \sup_{f \in F} \frac{e_{V_m, \|\cdot\|_\infty}(f)}{\|f\|_F}.$$

We can understand this assumption as follows: there is a sensible polynomial hierarchy  $V_m$  whose size grows like  $m^\sigma$  and for which the best weighted  $L^p$  approximation error of any  $f \in F$  decays like  $m^{-\alpha}$  once normalised by  $\|f\|_F$ .

**A3** (Sample work). The work required for a single evaluation of  $f_n$  satisfies

$$\text{Work}(f_n) \lesssim n^\gamma$$

for some  $\gamma > 0$ .

#### 11.4.2 Multi-Level Estimator:

We let  $h_l \rightarrow 0$  (which authors take to be the finite element of diameter  $l^{-1}$  but could in principle also be time step of an SDE) in the limit  $l \rightarrow \infty$  and consider a sequence of functions  $f_{h_0}, \dots, f_{h_l}$  which approximate in some suitable sense a function  $f$ . We then write  $f$  as a telescoping sum:

$$f = f_{h_0} + \sum_{l=1}^{\infty} (f_{h_l} - f_{h_{l-1}}) \quad (124)$$

As before, we cut-off this sum at some suitable  $L$ . Note that the coarsest approximation of  $f$ ,  $f_{h_0}$ , is taken to be of the same order as  $f$  and it is cheap to evaluate many samples. Further,  $(f_{h_l} - f_{h_{l-1}}) \rightarrow 0$  in the limit  $l \rightarrow \infty$  and samples are (increasingly) expensive to evaluate.

Choosing that  $(h_l)_{l=1}^\infty$  is exponentially decreasing and our sequence of subspaces  $(V_l)_{l=1}^\infty$  to be exponentially increasing we can write:

$$S_L(f) := \Pi_{V_L} f_0 + \sum_{l=1}^L \Pi_{V_{L-l}} (f_l - f_{l-1}) \quad (125)$$

where the operator  $\Pi_{V_l} : F \rightarrow V_l$ , for  $0 \leq l \leq L$ , is the random weighted least square approximation which projects onto the subspace  $V_l$ . *Because the projection at each level differs this sum does not telescope in the same way as (124), nor is it required to do so (all that is required is that the polynomial approximability assumption is fulfilled). This entails we get error contributions from each level  $l$ . However, in the asymptotic limit  $l \rightarrow \infty$   $V_l$  goes to infinity and  $f_l - f_{l-1} \rightarrow 0$ . This entails that our error goes to zero exponentially fast.*

#### 11.4.3 Multi-Level Convergence Analysis

##### Multi-Level Convergence Analysis

If  $\|f - f_h\|_F \lesssim h^\beta$  and each evaluation of  $f_h$  requires work  $h^{-\gamma}$ ,  $\forall h > 0$  for some space  $F$  for which  $\inf_{v \in V_l} \|g - v\|_{L^\infty(\Gamma)} \lesssim \|g\|_F (\dim V_l)^{-\alpha}$ ,  $\forall g \in F$ ,  $l \in \mathbb{N}$ , then the multilevel polynomial approximation (with optimal sampling) requires the work

$$W \simeq \varepsilon^{-\max\{\gamma/\beta, 1/\alpha\}} |\log \varepsilon|^r,$$

(with some  $r = r(\alpha, \beta, \gamma) > 0$  whose value is known) to achieve

- *Convergence in probability:*

$$\|S_L(f) - f\|_{L^2_\mu(\Gamma)} \leq \varepsilon \quad (126)$$

with probability  $> 1 - \varepsilon^{|\log \varepsilon|}$ ,  $0 < \varepsilon \ll 1$ .

- *Mean square convergence* (if  $L^\infty$  bound is available):

$$\mathbb{E} \|\tilde{S}_L(f) - f\|_{L^2_\mu(\Gamma)}^2 \leq \varepsilon^2.$$

Here we follow [2] in considering the particular case where  $\alpha, \gamma, \beta = 1$  and so we aim to show that  $W \simeq \epsilon^{-1} |\log(\epsilon)|^r$ , for some  $r$  depending on  $\alpha, \beta, \gamma$

*Proof.* First we derive an error bound. For any  $L > 0$  we have

$$\begin{aligned} \|S_L(f) - f_{h_L}\|_{L^2_\mu} &= \|\Pi_{V_L} f_0 + \sum_{l=1}^L \Pi_{V_{L-l}}(f_l - f_{l-1}) - \left(f_0 + \sum_{l=1}^L (f_l - f_{l-1})\right)\|_{L^2_\mu} \\ &= \|(\Pi_{V_L} - \text{Id}) f_0 + \sum_{l=1}^L (\Pi_{V_{L-l}} - \text{Id})(f_l - f_{l-1})\|_{L^2_\mu} \\ &\leq \underbrace{\|\Pi_{V_L} - \text{Id}\|_{F \rightarrow L^2}}_{\text{operator norm}} \|f_0\|_F + \sum_{l=1}^L \|\Pi_{V_{L-l}} - \text{Id}\|_{F \rightarrow L^2} \|f_l - f_{l-1}\|_F \\ &\leq \|\Pi_{V_L} - \text{Id}\|_{F \rightarrow L^2} \|f_0\|_F + \sum_{l=1}^L \|\Pi_{V_{L-l}} - \text{Id}\|_{F \rightarrow L^2} \|f_l - f\|_F. \end{aligned}$$

Assuming  $\dim V_l = e^l$  and  $h_l = e^{-l}$ , one has  $\|\Pi_{V_l} - \text{Id}\| \approx e^{-l} \approx \|f_l - f\|_F$ . Hence

$$\begin{aligned} \|S_L(f) - f\| &\leq \|S_L(f) - f_{h_L}\| + \|f_{h_L} - f\| \\ &\lesssim e^{-L} + \sum_{l=1}^L e^{-(L-l)} e^{-l} + e^{-L} = e^{-L} (L + 2). \end{aligned}$$

We note that in the asymptotic limit  $L \rightarrow \infty$  the error goes to 0. Next we derive the computational complexity of the Multi-Level Polynomial:

$$\begin{aligned} \text{Work}(S_L(f)) &= \text{Work}(\Pi_{V_L} f_0) + \sum_{l=1}^L \text{Work}(\Pi_{V_{L-l}}(f_l - f_{l-1})) \\ &\approx \dim V_L \log(\dim V_L) \text{Work}(f_0) + \sum_{l=1}^L \dim V_{L-l} \log(\dim V_{L-l}) \text{Work}(f_l - f_{l-1}) \\ &\lesssim e^L L + \sum_{l=1}^L e^{L-l} (L-l) e^l \\ &\approx e^L (L+1)^2. \end{aligned}$$

Denoting the right-hand side of the error bound  $e^{-L}(L+2) = \epsilon$ , we may rephrase the work bound as

$$\text{Work}(S_L(f)) \lesssim \varepsilon^{-1} |\log \varepsilon|^3 = \varepsilon^{-\max\{\gamma/\beta, 1/\alpha\}} |\log \varepsilon|^3.$$

□

## 11.5 Application of Multi-Level Least Squares Polynomial Regression Method to SDEs:

### 11.5.1 Original Example: Random PDEs

Let  $f(\mathbf{y}) = Q(u_{\mathbf{y}})$ , where  $u_{\mathbf{y}}$  is the solution to a PDE with random parameters  $\mathbf{y}$ . For example:

$$\begin{aligned} -\nabla \cdot (a(x, \mathbf{y}) \nabla u(x, \mathbf{y})) &= g(x) \quad \text{in } U \subset \mathbb{R}^D, \\ u(x, \mathbf{y}) &= 0 \quad \text{on } \partial U, \end{aligned} \tag{127}$$

with  $a : U \times \Gamma \rightarrow \mathbb{R}$  and  $\Gamma := [0, 1]^d$ .

We consider the case where  $f(\mathbf{y})$  cannot be evaluated exactly because we do not have access to an analytic solution of the PDE. In this case, we solve the PDE with finite difference or finite element solver giving approximations of  $f$ :  $f_n(\mathbf{y}) = Q(u_{\mathbf{y},n})$ . The general methodology to apply multilevel least squares regression then proceeds as follows:

1. Solve the PDE with the coarsest resolution  $h_0$  for a large number  $|\Gamma_L|$  of randomly chosen values of the parameters  $\mathbf{y}$ .
2. Extend these results to the entire parameter domain by means of a weighted least squares approximation in the large polynomial subspace  $V_L$ .
3. Solve the PDE with a finer resolution  $h_1$  and for the coarsest resolution  $h_0$  for a smaller number  $|\Gamma_{L-1}|$  of parameters  $\mathbf{y}$ .
4. Project the difference between coarse and fine resolution solutions onto the smaller polynomial subspace  $V_{L-1}$ .
5. Repeat for all levels  $l$ .
6. Sum all of the projections to obtain the final multi-level estimator for the function  $f$ .

**Of critical importance is that the samples  $\mathbf{y}$  used to compute  $f_l$  and  $f_{l-1}$  at a given level are the same. This guarantees that the sum (124) telescopes and our assumptions for the multi-level approximation are upheld.**

### 11.5.2 Application to SDEs: Initial issues

We want to estimate:

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}[\vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1 | \vec{P}_1 \cdot \vec{X}(t) = s, \vec{X}(0) = \vec{x}_0] \tag{128}$$

Where in the Black-Scholes model we have:

$$\vec{b}(t, X(t))_{ij} = X_i \Sigma_{ij} \tag{129}$$

Using the best mean square approximation we can relate finding an expectation value to an  $\mathcal{L}^2$  regression problem:

$$\begin{aligned} \mathbb{E}[X|Y] &= g(Y) \\ g(Y) &= \arg \min_{h \in V} \mathbb{E}[(X - h(Y))^2], \quad V = \{h : \mathbb{E}[h(Y)^2] < \infty\} \end{aligned} \tag{130}$$

**Single-Level:**

Model coefficients in time and space domain:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \int_0^T \mathbb{E} \left[ \left( \vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} - h(t, \vec{P} \cdot \vec{X}(t)) \right)^2 \mid \vec{X}(0) = \vec{x}_0 \right] dt \quad (131)$$

And then apply a Monte Carlo estimator:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M} \sum_m \frac{1}{N} \sum_n \left( \vec{P}^T b(t_n, \vec{X}^m(t_n)) b(t_n, \vec{X}^m(t_n))^T \vec{P} - h(t_n, \vec{P} \cdot \vec{X}^m(t_n)) \right)^2 \quad (132)$$

where  $\vec{X}^m(t)$  are independent realisations of  $\vec{X}$ . For ease of notation, define  $g := \vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1$ .

$$\mathcal{S}_L(f) := \Pi_{V_L} g_0(\vec{X}^0) + \sum_{l=1}^L \Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) \quad (133)$$

where

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \|(g(\vec{X}^l) - g(\vec{X}^{l-1})) - v(\cdot)\| \quad (134)$$

where we have purposefully left the argument(s) of  $v$  blank. The key difference between the case of random PDEs and this SDE case is that the data is a function of **two** different parameters  $\vec{X}^l$  and  $\vec{X}^{l-1}$ .

This gives us (at least) two options:

1. We **assume** that there exists a map  $\theta_{\Delta t} : \vec{X}^l \rightarrow \vec{X}^{l-1}$  such that in the limit  $\Delta t \rightarrow 0$ ,  $\vec{X}^{l-1} = \vec{X}^l$ . Then we can write:

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \|(g(\vec{X}^l) - g(\theta_{\Delta t}(\vec{X}^l))) - v(\vec{X}^l)\| \quad (135)$$

This map will be dependent on the particular path in question:  $\theta_{\Delta t}(\vec{X}^l, \omega)$ . The question is whether such a map exists and whether its properties allow for a telescoping sum to be written.

2. We allow that  $v$  is a function of two variables  $v = v(\vec{X}^{l-1}, \vec{X}^l)$ . Then we can write:

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \|(g(\vec{X}^l) - g(\vec{X}^{l-1})) - v(\vec{X}^{l-1}, \vec{X}^l)\| \quad (136)$$

Since the function to be estimate  $\bar{b}^2(t, s)$  is a function of only one variable and assuming that  $X_l$  and  $X_{l-1}$  are close, we stipulate that  $\bar{b}^2(t, s) = v(s, s)$ . That is, we evaluate  $v$  along the diagonal. Important considerations will be i) determining an appropriate ansatz for  $v$  and ii) limiting the computational cost of doubling the dimension. We note that on the coarsest level, the assumption that  $X_l$  and  $X_{l-1}$  are close will be most strained since this is when the coarse and fine paths are furthest apart.

### 11.5.3 Application to SDEs: Option 1

We need:

1. Sequence of exponentially increasing polynomial subspaces  $(V_l)_{l=0}^\infty$ : We chose subspaces of (total degree) Legendre polynomials which are orthogonal on the interval  $[-1, 1]$  with weight function  $w = 1$ .
2. Exponentially decreasing time steps  $h_0, h_1, \dots, h_l, \dots$ : We chose  $h_l = h_0 \cdot 2^{-l}$ .

We simulate  $M_l$  fine paths  $X^f(t^l)$  and construct the coarse paths  $X^c(t^{l-1})$  by summing the Brownian increments of the fine paths. We then have a choice:

1. Interpolate the coarse path on the missing fine time steps using a Brownian Bridge.

2. Regress the difference only on the coarse time intervals.

We assumed that the option 2 is sufficient. Further, we chose that  $h = h(t_n, \vec{P} \cdot \vec{X}_f^m(t_n^c))$ , so  $h$  was a function of the fine paths. For  $l > 0$ , the Multilevel Monte Carlo least squares estimator for the projected diffusion coefficient can then be written as follows:

$$(\bar{b}_l^{x_0})^2(\cdot, \cdot) - (\bar{b}_{l-1}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M_l} \sum_{m=1}^{M_l} \frac{1}{N_{l-1}} \sum_{n=1}^{N_{l-1}} \left( [\vec{P}^T b(t_n, \vec{X}_l^m(t_n^{l-1})) b(t_n, \vec{X}_l^m(t_n^{l-1}))^T \vec{P} - \vec{P}^T b(t_n, \vec{X}_{l-1}^m(t_n^{l-1})) b(t_n, \vec{X}_{l-1}^m(t_n^{l-1}))^T \vec{P}] - h(t_n^{l-1}, \vec{P} \cdot \vec{X}_l^m(t_n^{l-1})) \right)^2 \quad (137)$$

#### 11.5.4 Application to SDEs: Optimal Transport Map

In some applications, one seeks to construct a *measure-preserving transformation* from a one-dimensional continuous random variable  $X$  to another random variable  $Y$ , such that the push-forward of the law of  $X$  under the map coincides with the law of  $Y$ . More precisely, we look for a measurable map

$$T : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{such that} \quad T_{\#} \mu_X = \mu_Y,$$

where  $\mu_X$  and  $\mu_Y$  denote the probability measures of  $X$  and  $Y$ , and  $T_{\#} \mu_X$  denotes the push-forward measure. Among the infinitely many maps that may satisfy this property, *optimal transport* (OT) theory selects the one that minimizes a given transportation cost functional. A common choice in the quadratic case is

$$T^* = \arg \min_{T_{\#} \mu_X = \mu_Y} \mathbb{E} [|T(X) - X|^2],$$

which, under mild regularity assumptions, admits a unique monotone non-decreasing solution in the one-dimensional setting [9–11].

In practice, when only empirical samples  $\{(x_i, y_i)\}_{i=1}^N$  are available, the OT map can be approximated via *discrete optimal transport algorithms*. These include the classical *Hungarian algorithm* for the exact discrete assignment problem and modern *entropic regularization* techniques [12], which lead to the *Sinkhorn algorithm*—a scalable method widely used in machine learning and data-driven applications. For the one-dimensional case, the OT map simplifies to a *quantile map*

$$T^*(x) = F_Y^{-1}(F_X(x)),$$

where  $F_X$  and  $F_Y$  are the cumulative distribution functions (CDFs) of  $X$  and  $Y$ , respectively. This formula provides an explicit and computationally trivial way to transport samples from  $X$  to the law of  $Y$ .

# A Probability Theory

## A.1 Definitions

**Probability Space:** A probability space is a triple  $(\Omega, \mathcal{F}, \mathbb{P})$  where:

- $\Omega$ : sample space, the set of all possible outcomes;
- $\mathcal{F}$ :  $\sigma$ -algebra of events (subsets of  $\Omega$ );
- $\mathbb{P}$ : probability measure  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ .

**$\sigma$ -algebra:** A collection  $\mathcal{F}$  of subsets of  $\Omega$  is called a  $\sigma$ -algebra if:

1.  $\Omega \in \mathcal{F}$  and  $\emptyset \in \mathcal{F}$ ,
2. If  $A \in \mathcal{F}$ , then  $A^c = \Omega \setminus A \in \mathcal{F}$ ,
3. For any countable collection  $\{A_i\} \subset \mathcal{F}$ , we have  $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ .

One example of a  $\sigma$ -algebra is the **Borel  $\sigma$ -algebra**  $\mathcal{B}(\mathbb{R})$ . The Borel  $\sigma$ -algebra is the smallest  $\sigma$ -algebra on  $\mathbb{R}$  containing all open intervals. It includes:

$$(-\infty, a), \quad (b, \infty), \quad [a, b], \quad (a, b], \quad \{a\}, \quad \{a_1, \dots, a_n\}, \quad \text{etc.}$$

**Measure:** Let  $(\Omega, \mathcal{F})$  be a measurable space. A function  $\mu : \mathcal{F} \rightarrow [0, \infty]$  is a measure if:

- $\mu(\emptyset) = 0$ ,
- For disjoint sets  $F_1, F_2, \dots \in \mathcal{F}$ ,

$$\mu\left(\bigcup_{i=1}^{\infty} F_i\right) = \sum_{i=1}^{\infty} \mu(F_i).$$

If  $\mu(\Omega) = 1$ , then  $\mu$  is a probability measure. **Probability Measure:** Let  $(\Omega, \mathcal{F})$  be a measurable space. A probability measure  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$  satisfies:

1.  $\mathbb{P}(\Omega) = 1$ ,
2.  $\mathbb{P}(A) \geq 0$  for all  $A \in \mathcal{F}$ ,
3. For mutually disjoint sets  $A_1, A_2, \dots \in \mathcal{F}$ ,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The probability measure assigns a probability to each event in  $\mathcal{F}$ .

**$\mathcal{F}$ -measurable Function:** Let  $(\Omega, \mathcal{F})$  be a measurable space and  $f : \Omega \rightarrow \mathbb{R}$ . Then  $f$  is  $\mathcal{F}$ -measurable if:

$$f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\} \in \mathcal{F}, \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

Equivalently:

$$y \in \mathbb{R} \rightarrow \{\omega \in \Omega : f(\omega) \leq y\} \in \mathcal{F}, \quad \forall y \in \mathbb{R}.$$

Example: A random variable  $X_t$  is  $\mathcal{H}_t$ -measurable if the truth value of statements about  $X_t$  - e.g.  $X_t \leq 5$  can be determined using only the information contained in  $\mathcal{H}_t$ .

**Lebesgue integral:** Let  $(\Omega, \mathcal{F})$  be a measure space with measure  $\mu$ , and let  $f : \Omega \rightarrow \mathbb{R}$  be  $\mathcal{F}$ -measurable. Decompose  $f = f^+ - f^-$  with  $f^+(\omega) = \max\{f(\omega), 0\}$  and  $f^-(\omega) = \max\{-f(\omega), 0\}$ . Define

$$\int_{\Omega} f d\mu := \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu,$$

whenever at least one of the two integrals on the right-hand side is finite.

**Random variable:** On a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ , a *real-valued random variable* is an  $\mathcal{F}$ -measurable map  $X : \Omega \rightarrow \mathbb{R}$ . Thus, for any Borel set  $B \subset \mathbb{R}$ , the pre-image  $X^{-1}(B)$  lies in  $\mathcal{F}$ .

**Distribution function:** For a random variable  $X$ , its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(\{\omega : X(\omega) \leq x\}),$$

and for any  $a < b$ ,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

**Probability density:** A measurable function  $f : \mathbb{R} \rightarrow [0, \infty)$  with  $\int_{\mathbb{R}} f(x) dx = 1$  is called a (Lebesgue) *density*. A random variable  $X$  has density  $f$  if

$$\mathbb{P}(X \in A) = \int_A f(x) dx, \quad A \in \mathcal{B}(\mathbb{R}).$$

## A.2 Conditional Expectation

**Conditional Expectation:** Let  $X$  be an integrable random variable on the probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  and let  $\mathcal{G} \subset \mathcal{F}$  be a  $\sigma$ -algebra. The conditional expectation of  $X$  given  $\mathcal{G}$  is a random variable

$$\mathbb{E}[X \mid \mathcal{G}]$$

such that

1.  $\mathbb{E}[X \mid \mathcal{G}]$  is  $\mathcal{G}$ -measurable;
2. for every  $A \in \mathcal{G}$ ,

$$\mathbb{E}[\mathbf{1}_A \mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[\mathbf{1}_A X].$$

The random variable  $\mathbb{E}[X \mid \mathcal{G}]$  satisfying (i)–(ii) is essentially unique: any two versions differ only on a set in  $\mathcal{G}$  of  $\mathbb{P}$ -measure zero.

**Observe.** Taking  $A = \Omega$  in (ii) gives

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X].$$

**Properties of conditional expectation of random variables:** Let  $X$  and  $Z$  be integrable random variables on  $(\Omega, \mathcal{F}, \mathbb{P})$  and let  $\mathcal{G} \subset \mathcal{F}$  be a  $\sigma$ -algebra.

1. **Linearity:**

$$\mathbb{E}[\alpha X + Z \mid \mathcal{G}] = \alpha \mathbb{E}[X \mid \mathcal{G}] + \mathbb{E}[Z \mid \mathcal{G}], \quad \text{for all } \alpha \in \mathbb{R}.$$

2. **Monotonicity:** If  $X \geq Z$  almost surely, then

$$\mathbb{E}[X \mid \mathcal{G}] \geq \mathbb{E}[Z \mid \mathcal{G}].$$



3. **Multiplication by a measurable function:** If  $\mathbb{E}[|XZ|] < \infty$  and  $Z$  is  $\mathcal{G}$ -measurable, then

$$\mathbb{E}[XZ \mid \mathcal{G}] = Z \mathbb{E}[X \mid \mathcal{G}], \quad \text{and} \quad \mathbb{E}[Z \mid \mathcal{G}] = Z.$$

4. **Independence:** If  $X$  is independent of  $\mathcal{G}$ , then

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X].$$

5. **Tower property (iterated expectation):** If  $\mathcal{H} \subset \mathcal{G} \subset \mathcal{F}$ , then

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{H}].$$

6. **Best predictor in  $L^2$ :** If  $Z$  is any  $\mathcal{G}$ -measurable random variable with  $\mathbb{E}[Z^2] < \infty$ , then

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])^2] \leq \mathbb{E}[(X - Z)^2].$$

7. **Jensen's inequality:** Let  $\varphi : \mathbb{R} \rightarrow \mathbb{R}$  be a convex function such that  $\mathbb{E}[\varphi(X)] < \infty$ . Then

$$\varphi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}].$$

### A.3 Continuity

**Hölder continuity:** A real or complex-valued function  $f$  on  $d$ -dimensional Euclidean space is Hölder continuous if there exist real constants  $C > 0$ ,  $\alpha > 0$ , such that for any  $x$  and  $y$  in the domain of  $f$

$$|f(x) - f(y)| \leq C \|x - y\|^\alpha.$$

$\alpha$  is called the *exponent of the Hölder continuity*.

**Lipschitz continuity:** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is Lipschitz continuous if  $\exists K \in \mathbb{R}^+$  such that for  $x, y \in \mathbb{R}$ :

$$|f(x) - f(y)| \leq K|x - y|. \quad (138)$$

If  $a(X_t, t)$  and  $b(X_t, t)$  are Lipschitz continuous, then a solution to the stochastic differential equation

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad X_0 = x_0 \quad (139)$$

exists.

**Continuity:**  $\forall \epsilon \exists \delta$  such that  $|f(x) - f(y)| < \epsilon$  if  $|x - y| < \delta$ . Lipschitz continuity implies continuity.

### A.4 Convergence

**Weak form of Lax Equivalence:** (Loosely) The combination of consistence and stability imply convergence. Stability means that your function is bounded in some sense and consistence means that the global error tends to zero.

**Strong Convergence:** The sequence of random variables  $\{Y_n\}_{n \in \mathbb{N}}$  converges strongly to the random variable  $Y$  if:

$$\|Y - Y_n\|_{L^2(\Omega)} \equiv \sqrt{\mathbb{E}[(Y - Y_n)^2]} \rightarrow 0 \quad (140)$$

By Chebychev's inequality we have that for any fixed  $\epsilon > 0$ :

$$P(|Y - Y_n| \geq \epsilon) \leq \frac{\mathbb{E}[(Y - Y_n)^2]}{\epsilon^2} \rightarrow 0 \quad (141)$$

**Weak Convergence:** The sequence of random variables  $\{Y_n\}_{n \in \mathbb{N}}$  converges weakly to the random variable  $Y$  if for all bounded continuous function  $g$ :

$$\mathbb{E}[g(Y)] - \mathbb{E}[g(Y_n)] \rightarrow 0 \quad (142)$$

Strong convergence implies weak convergence but not vice versa in general.

**Convergence in probability:** Let  $(X_n)_{n \geq 1}$  and  $X$  be random variables defined on the same probability space. We say that  $X_n$  converges to  $X$  *in probability*, written

$$X_n \xrightarrow{P} X,$$

if for every  $\varepsilon > 0$ ,

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0.$$

**Convergence in distribution:** We say that  $X_n$  converges to  $X$  *in distribution* written

$$X_n \xrightarrow{d} X,$$

if the cumulative distribution functions  $F_{X_n}(x)$  satisfy

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad \text{for all } x \text{ at which } F_X \text{ is continuous.}$$

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